

**INTELLIGENT CAR RECOMMENDATION AND PRICING
SYSTEM USING MACHINE LEARNING**

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**THIS THESIS IS SUMMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF
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DEVELOPMENT)WITH HONOURS**

**FACULTY OF INFORMATICS AND COMPUTING
UNIVERSITI SULTAN ZAINAL ABIDIN**

2026

DECLARATION

I hereby declare that the report is based on my original work except for quotations and citations, which have been duly acknowledged. I also declare that it has not been previously or concurrently submitted for any other degree at Universiti Sultan Zainal Abidin or other institutions.

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CONFIRMATION

This is to confirm that this project entitled "Intelligent Car Recommendation and Pricing System Using Machine Learning" was prepared and submitted by Tuan Mifzal Fawwaz Bin Tuan Fauzi (Matric Number: [REDACTED]) and has been found satisfactory in terms of scope, quality, and presentation as partial fulfillment of the requirements for the Bachelor of Computer Science (Software Development) with Honours at Universiti Sultan Zainal Abidin.

The research conducted and the writing of this report were carried out under my supervision.

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Date : 15 JULY 2026

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
AI	ARTIFICIAL INTELLIGENCE
ML	MACHINE LEARNING
FYP	FINAL YEAR PROJECT
DFD	DATA FLOW DIAGRAM
ERD	ENTITY RELATIONSHIP DIAGRAM
UI	USER INTERFACE
UX	USER EXPERIENCE
API	APPLICATION PROGRAMMING INTERFACE

DB	DATABASE
SQL	STRUCTURED QUERY LANGUAGE
RF	RANDOM FOREST
XGBOOST	EXTREME GRADIENT BOOSTING
KNN	K-NEAREST NEIGHBOUR
SVM	SUPPORT VECTOR MACHINE
LR	LINEAR REGRESSION
R^2	COEFFICIENT OF DETERMINATION
RMSE	ROOT MEAN SQUARED ERROR
MAE	MEAN ABSOLUTE ERROR

MSE	MEAN SQUARED ERROR
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CSV	COMMA-SEPARATED VALUES
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HTTP	HYPERTEXT TRANSFER PROTOCOL
------	-----------------------------

JSON	JAVASCRIPT OBJECT NOTATION
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IDE	INTEGRATED DEVELOPMENT ENVIRONMENT
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CRUD	CREATE, READ, UPDATE, DELETE
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ABSTRACT

The rapid growth of online automotive marketplaces has increased the number of vehicle listings available to consumers, making it difficult for buyers to identify suitable vehicles and for sellers to determine competitive selling prices. Most existing car sales platforms rely on static search filters and manual browsing, which do not consider user preferences, browsing behaviour, or historical interactions. In addition, vehicle prices are commonly determined based on seller judgment rather than data-driven analysis, leading to inconsistent pricing and reduced buyer confidence.

This project proposes an **Intelligent Car Recommendation and Pricing System Using Machine Learning** to improve decision-making for both buyers and sellers.

The proposed web-based system integrates two machine learning approaches:

Collaborative Filtering to generate personalised vehicle recommendations based on user preferences and interaction history, and **Random Forest Regression** to predict fair market prices using vehicle attributes such as brand, model, manufacturing year, mileage, engine type, and vehicle condition. The system consists of three user modules: Buyer, Seller, and Administrator, each providing functionalities that support efficient vehicle management and decision-making.

The proposed system was evaluated based on recommendation effectiveness, pricing prediction reliability, and overall system usability. The results demonstrate that integrating personalised recommendations with machine learning-based price prediction improves recommendation relevance, increases pricing transparency, and assists users in making more informed purchasing and selling decisions. Overall, the system provides an intelligent decision-support platform that enhances user experience and contributes to the development of more efficient online automotive marketplaces.

ABSTRAK

Pertumbuhan pesat platform jual beli kenderaan secara dalam talian telah meningkatkan jumlah pilihan kenderaan yang tersedia kepada pengguna, sekali gus menyukarkan pembeli untuk mengenal pasti kenderaan yang sesuai serta penjual untuk menetapkan harga jualan yang kompetitif. Kebanyakan platform jualan kereta sedia ada masih bergantung kepada penapisan carian statik dan pencarian secara manual yang tidak mengambil kira keutamaan pengguna, tingkah laku pelayaran, atau sejarah interaksi pengguna. Selain itu, harga kenderaan lazimnya ditentukan berdasarkan pertimbangan penjual tanpa analisis berasaskan data, yang boleh menyebabkan ketidakselarasan harga dan mengurangkan keyakinan pembeli.

*Projek ini mencadangkan **Sistem Cadangan dan Penetapan Harga Kereta Pintar Menggunakan Pembelajaran Mesin** bagi meningkatkan proses membuat keputusan untuk pembeli dan penjual. Sistem berasaskan web yang dicadangkan ini menggabungkan dua pendekatan pembelajaran mesin, iaitu **Collaborative Filtering** untuk menghasilkan cadangan kenderaan yang diperibadikan berdasarkan keutamaan pengguna dan sejarah interaksi, serta **Random Forest Regression** untuk meramalkan harga pasaran yang munasabah berdasarkan atribut kenderaan seperti jenama, model, tahun pembuatan, perbatuan, jenis enjin dan keadaan kenderaan. Sistem ini menyokong tiga jenis pengguna, iaitu pembeli, penjual dan pentadbir, dengan fungsi yang direka bagi membantu pengurusan kenderaan serta menyokong proses membuat keputusan dengan lebih berkesan.*

Sistem yang dicadangkan telah dinilai berdasarkan keberkesanan cadangan, kebolehpercayaan ramalan harga dan kebolehgunaan sistem secara keseluruhan. Hasil penilaian menunjukkan bahawa penggabungan cadangan yang diperibadikan dengan ramalan harga berasaskan pembelajaran mesin dapat meningkatkan kerelevanan cadangan, meningkatkan ketelusan penetapan harga serta membantu pengguna membuat keputusan pembelian dan penjualan dengan lebih yakin. Secara keseluruhannya, sistem ini menyediakan platform sokongan keputusan yang pintar; meningkatkan pengalaman pengguna dan menyumbang kepada pembangunan platform jual beli kenderaan dalam talian yang lebih cekap.

CHAPTER 1

1.1 Introduction

There has been a lot of transformation in the automotive sector with the growth of information technology and machine learning. Online car sales platforms have gained popularity due to convenience and ease. Nevertheless, the proliferation of car listings has created information overload and thus complicated the ability of car users to discern which vehicles fit their preferences and budget the most. Traditional car sales platforms mostly use static search filters, such as brand, price range and vehicle type. These methods cannot analyse user behaviour or market trends, which is the case with given recommendations that are usually of a generic nature and not very accurate pricing. Machine learning is an effective way to solve this problem by making systems learn from historical data and suggest intelligent recommendations and prices. This project presents an Intelligent Car Recommendation and Pricing System Using Machine Learning which will improve the decision-making process of buyers while helping sellers to price their cars in a competitive manner.

1.2 Project Background

The technological development and application of cars has changed dramatically thanks to rapid development of information technology and machine learning technology, especially in the online car sales field. With growing use of digital marketplaces, consumers can now find, compare and buy vehicles through web-based systems without being constrained in their shopping. These platforms provide access to a wide range of vehicle listings, providing convenience and market reach for both buyers and sellers.

Despite these benefits, most current online car sales platforms still use traditional listing-based strategies and rudimentary filtering systems. Users are usually obliged to manually search and filter vehicles according to narrow criteria such as brand, price range and location. This approach is often resulting in information overload and it is difficult for buyers to find vehicles that are best matched to their preferences, budget and usage needs. Concurrently, sellers have been encountering challenges in deciding competitive and fair prices owing to the lack of data-driven pricing tools.

Machine learning offers an exciting way to solve these problems by allowing systems to learn patterns from past data, and how people interact with systems.

Recommendation systems can be used to analyse user preferences and behaviour and provide personalised recommendations on vehicles, while predictive models may be used to estimate the price of a car by analysing relationships between vehicle attributes such as brand, year of manufacture, mileage and engine specifications. These intelligent techniques improve the accuracy of decision making and reduce the level of uncertainty of the car buying and selling process.

This project proposes an Intelligent Car Recommendation and Pricing System Using Machine Learning in order to improve the functionality of online car sales platforms. Using a combination of collaborative filtering for personalised recommendation and a regression based models for price prediction, the proposed system is expected to help a buyer to choose suitable vehicles and help a seller to price their vehicle better. The proposed system is about creating a better user experience and pricing transparency and overall efficiency in the digital automotive marketplaces.

1.3 Problem Statement

The accelerated development of online car sales platforms has led to the accessibility of information related to vehicles and the widening of the market, both for buyers and sellers. However, despite the availability of large volumes of data, there are still a substantial number of existing data platforms that lack intelligent mechanisms to support effective decision-making activities. Most systems are based on static listing and simple filtering capabilities forcing the user to manually browse many vehicle choices. This process is time consuming, inefficient, and overwhelming for buyers in many cases.

Buyers often have problems finding vehicles to use according to their personal preferences, budget limits, and use requirements. The lack of personalised recommendation features produces generic search results which is not reflective of individuals user needs. As a result, users can express uncertainty and dissatisfaction in the vehicle selection process.

In addition, pricing of cars is a major challenge in the online marketplaces. Sellers usually determine the price according to their own judgement or climate observation that might not be an accurate calculation of the fair market value of a car. Buyers, on the other hand, often have little confidence in judging whether the listed price is reasonable, competitive and/or inflated. The lack of data-driven price prediction tools is a factor in the lack of transparency and trust between buyers and sellers.

Although machine learning techniques have proven to be effective in recommendation systems and predictive analytics, in many current car sales platforms both personalised recommendations and intelligent pricing are not fully utilised within a single platform. This disparity shows how there is need for an intelligent solution that is able to analyse user behaviour and vehicle data to enhance the accuracy of recommendations and the reliability of pricing.

Therefore, this project intends to solve these problems by designing an Intelligent Car Recommendation and Pricing System Using Machine Learning that supports personalised recommendations of cars and reliable prediction of pricing for improving user experience, decision-making efficiency, and trust in online car sales platforms.

1.4 Objectives

The main objective of this project is to create an intelligent system that will enrich the car buying and selling process using the machine learning techniques. Specifically, the objectives follow:

1. To build a smart car recommendation system that can provide tailored vehicle recommendations based on user preferences, browsing history and interaction history.
2. To create a machine learning-based car pricing prediction model to estimate fair market values using relevant characteristics of the car (e.g., brand, year of manufacture, mileage and specifications).
3. To examine the performance of the proposed system in terms of recommendation accuracy, pricing reliability and usability in general.

1.5 Scope

The scope of this project defines the functional boundaries and coverage of the Intelligent Car Recommendation and Pricing System Using Machine Learning project.

The system is envisioned as a web-based system which focuses on offering intelligent decision support for online car buying and selling activities. The scope is divided into three main user modules: Buyer, Seller and Administrator.

1.5.1 Buyer

The Buyer module is concerned with helping users to find an appropriate vehicle and to make sense of price information. The buyers have to register and login to the system before accessing any features of the system. Once the buyer is authenticated, they can control information on their personal profile and be able to go through the available car listings provided by sellers.

Buyers can search for vehicles according to certain criteria and get personalised recommendations for cars based on the machine learning model of recommendations. These recommendations are generated based on buyer preferences and history of interaction. Additionally, buyers can access predicted car prices for comparison purposes to check if a listed price is reasonable or not. The system offers aggregated results also which consolidates the recommendation ranking, predicted price and cars availability information to support decision making.

1.5.2 Seller

The Seller module is created to cater to individuals or dealers who want to list cars for sale on the platform. Sellers are asked to register and log in to the system in order to access the functionalities of sellers. Sellers are able to control their profile and add new car listings via the addition of specific vehicle attributes such as brand, model, year made, mileage and engine type.

Sellers may also make updates or remove existing listings for car sales as necessary. A major function of the Seller module is the price prediction function, in which sellers can enter the details of their car on the system and get the proposed price of the car using machine learning based pricing function. These predicted prices help sellers to make better and more competitive pricing decisions based on data-driven analysis.

1.5.3 Admin

Overall system management and monitoring responsibility is the role of the Administrator module. Administrators have the right to control user accounts for both buyers and sellers, and make sure that only legit users are able to access the platform. Administrators also monitor the listings of cars to ensure the rules of the system and the validity of the data.

Furthermore, administrators monitor system integrity by validating stored data and tracking system activity for maintenance such purposes. The administrator can view system data to aid in monitoring, maintenance and performance evaluation of the platform.

1.5.4 System and Technical Scope

From the system perspective, the developed project focuses on the creation of a web-based application that combines machine learning models for car recommendation and prediction. The recommendation module uses the collaborative filtering techniques and the pricing module uses the regression-based models trained on the vehicle attributes.

The system does not cover online payment processing, vehicle inspection services or direct transaction handling between buyers and sellers. The scope is limited to decision support features for allowing to increase recommendation accuracy and transparency pricing.

1.6 Limitation of Work

Although the Intelligent Car Recommendation and Pricing System Using Machine Learning provides a helpful decision support for the online buying and selling of cars, some limitations are recognized.

Firstly, the quality of the recommendation for car and the prediction of the prices depends greatly on the quality and completeness of the data in use. Machine learning models are based on historic vehicle data along with user interaction information to

arrive at the predictions. In one situation, if the limited dataset is old or contains inconsistencies then the accuracy of recommendations and predicted price may suffer.

Secondly, the system suffers from the cold start issue where mainly the new buyers who have not come in contact with the platform. Since the recommendation model examines user preferences and interaction history, the recommendations for newly registered users may not be as personalised at first, until sufficient behavioural information is gathered.

Thirdly, the price prediction model is used to make estimates of car prices based on predefined vehicle attributes like brand of the car, model of the car, year of the car, mileage of the car, type of the engine. External factors are not fully modeled when there are abrupt changes in the market, economic conditions or the market demand for vehicles, incorrect approaches and promotion in selling strategy by different sellers or the enterprise. As a result, the predicted prices might not always be true of the real-time market conditions.

In addition to it, the scope of the system is restricted to decision support functionalities. The system is not integrated with online payment mechanisms, vehicle inspection services or direct transaction services conducted between the buyers and sellers. These features are outside the scope of this study and would require more technical, legal and security considerations.

Lastly, the system is designed in the form of a web-based application and tested in a controlled environment. Performance under high traffic conditions or large scale deployment has not been extensively tested, which may impact this in real-world use cases including system scalability.

1.7 Expected Result

The Intelligent Car Recommendation and Pricing System Using Machine Learning is expected to provide a number of measurable and practical results to improve the efficiency of online car sales Web sites.

Firstly, the system is able to offer a personalised recommendations of cars that will perfectly correspond to buyer preferences, browsing behaviour and history of interactions. By using collaborative filtering techniques, the recommendation module should be more efficient in terms of time and effort needed for buyers to find suitable vehicles than the manual searching techniques.

Secondly, the car price prediction module is expected to produce reasonable and data driven car prices based on the attributes of the car such as brand, model, year of manufacture, mileage and engine type. These predicted prices are foreseen to lead to more transparency of prices and help buyers make a judgment of the fairness of listed prices and help sellers set competitive and realistic price ranges.

Thirdly, the integration of the features for recommendations and pricing into one web-based platform is expected to bring the level of efficiency of decision making by users to a new height. Buyers can use the recommendations for vehicles to compare with expected prices, and sellers can make informed pricing decisions with the background of analytical models.

In addition, the system is expected to show the practical applicability of machine-learning techniques in automotive-e-commerce. Successful implementation of collaborative filtering and regression models will testify to the effectiveness of these models in real-world decision support systems.

Overall, the society may expect that this research will bring improved user satisfaction, more trust in online car marketplaces, and improved system functionality as compared to the traditional static listing platforms.

1.8 Activities, Milestones (Gantt Chart)

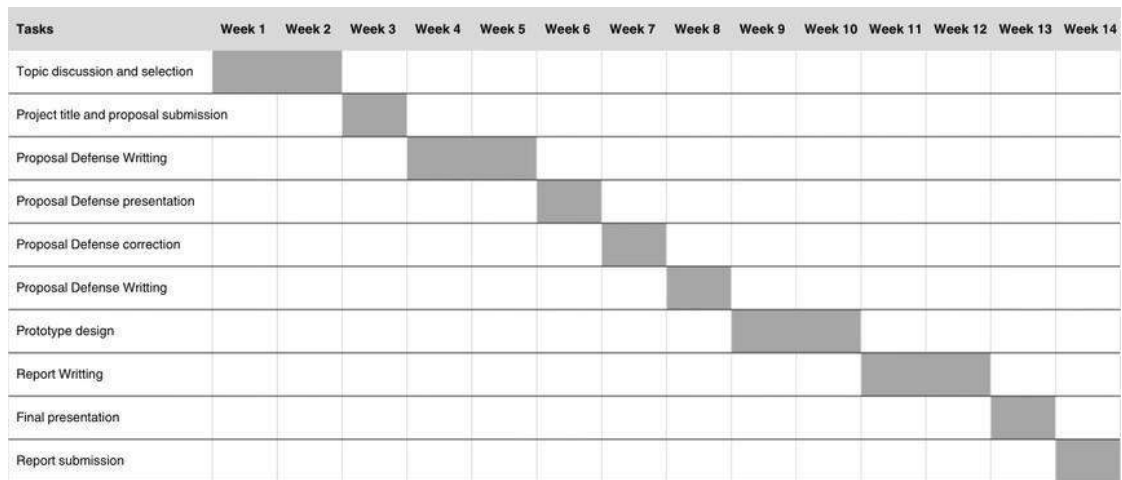


Figure 1.1:Gantt Chart

1.9 Summary of the Chapter

This chapter introduced Intelligent Car Recommendation and Pricing System Using Machine Learning by providing the background and motivation for the study. The challenges faced in existing online car sales platforms were discussed, which emphasize the problems related to lack of personalised recommendations and data driven pricing support.

The chapter defined problem statement, project objectives, scope of study, limitations, significance, expected results of the problem system. These sections altogether determine the purpose, boundaries, and importance of the project and set clear expectations of the project's outcomes.

In sum, this chapter forms a strong base of the research, by justifying why the need for an intelligent car recommendation and pricing system. The following chapter will cover the related studies and available studies to further establish and support the design and development of the proposed system.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing theories, techniques, and studies related to intelligent recommendation systems, machine learning-based pricing models, and online car marketplaces. The purpose of this literature review is to examine relevant algorithms, technological approaches, and existing systems in order to identify strengths, limitations, and research gaps. The findings from this chapter provide a theoretical foundation and justify the design choices made for the proposed **Intelligent Car Recommendation and Pricing System Using Machine Learning**.

2.2 Technique

This section talks about the basic theories and techniques involved with recommendation system and pricing prediction that apply to the proposed system.

2.2.1 Hybrid Filtering Algorithm

Hybrid filtering algorithms combine different recommendation paradigms, usually the content-based filtering paradigm and the collaborative filtering paradigm, to improve the accuracy of the recommendations. This method adds some solutions to common limitations, such as the cold-start problem and data sparsity. Hybrid models are already widely used in e-Commerce platforms, to provide more reliable and personalised recommendations.

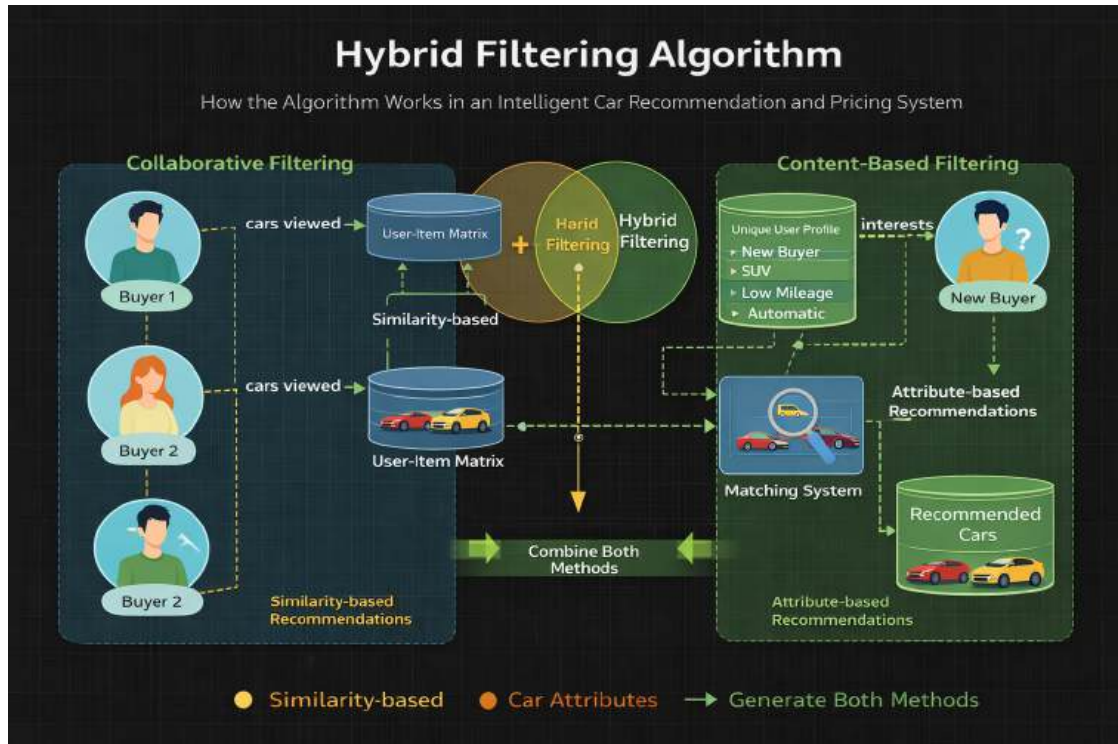


Figure 2.1: Hybrid Filtering Algorithm

In the automotive space, the combination of vehicle attributes and user interaction patterns using hybrid filtering can create more accurate car recommendations.

2.2.2 Content-Based Filtering Algorithm

Content-based filtering suggests content based on similarity between the features of items and users preferences. In the case of a car recommendation system, this approach uses data attributes of vehicles such as brand, model, engine type and price range and matches with user-selected preferences. Although content based filtering works well for users that are new to using a system, it might possibly be limiting in the diversity of recommendations because it is primarily looking for recommendations related to items which the user has previously viewed or selected in the past.

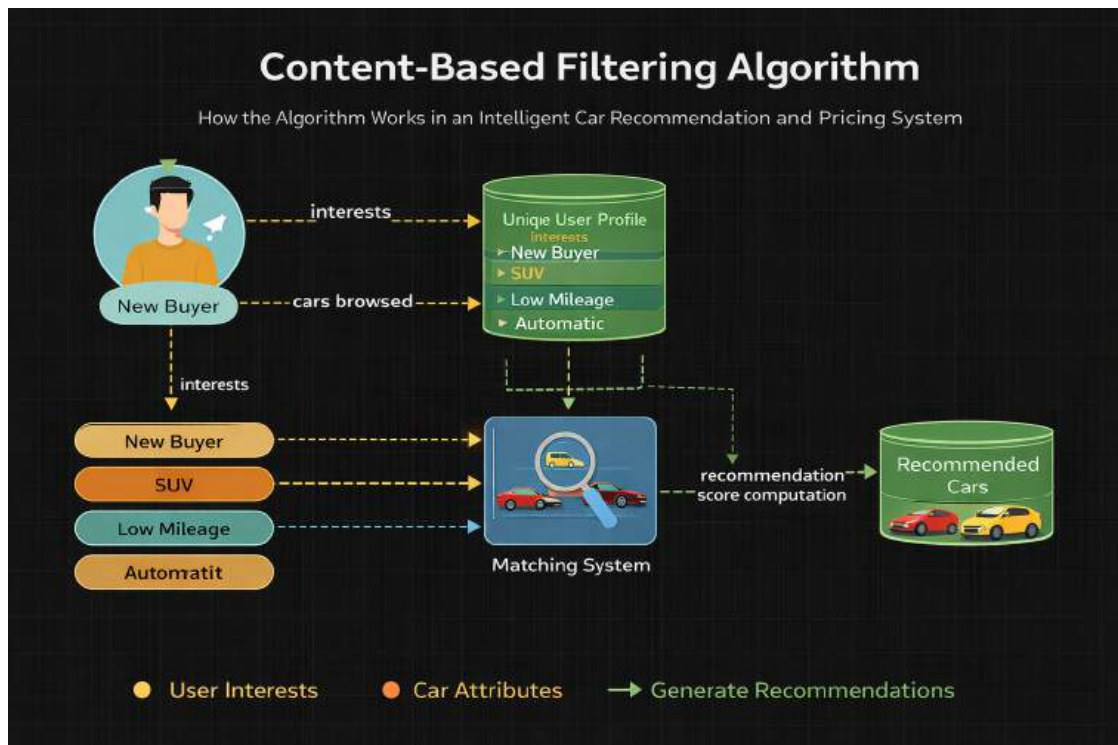


Figure 2.2 Content-Based Filtering Algorithm

2.2.3 Collaborative Filtering Algorithm

Collaborative filtering produces recommendations based on analysing user behaviour and interactions, e.g. viewed cars or selected preferences. This way of thinking is based on the assumption that the users with similar interest in the past will have similar preferences in the future. Collaborative filtering is good practice in giving personalized recommendations and does not need an item description. However, it needs sufficient interaction data from users for it to perform optimally.

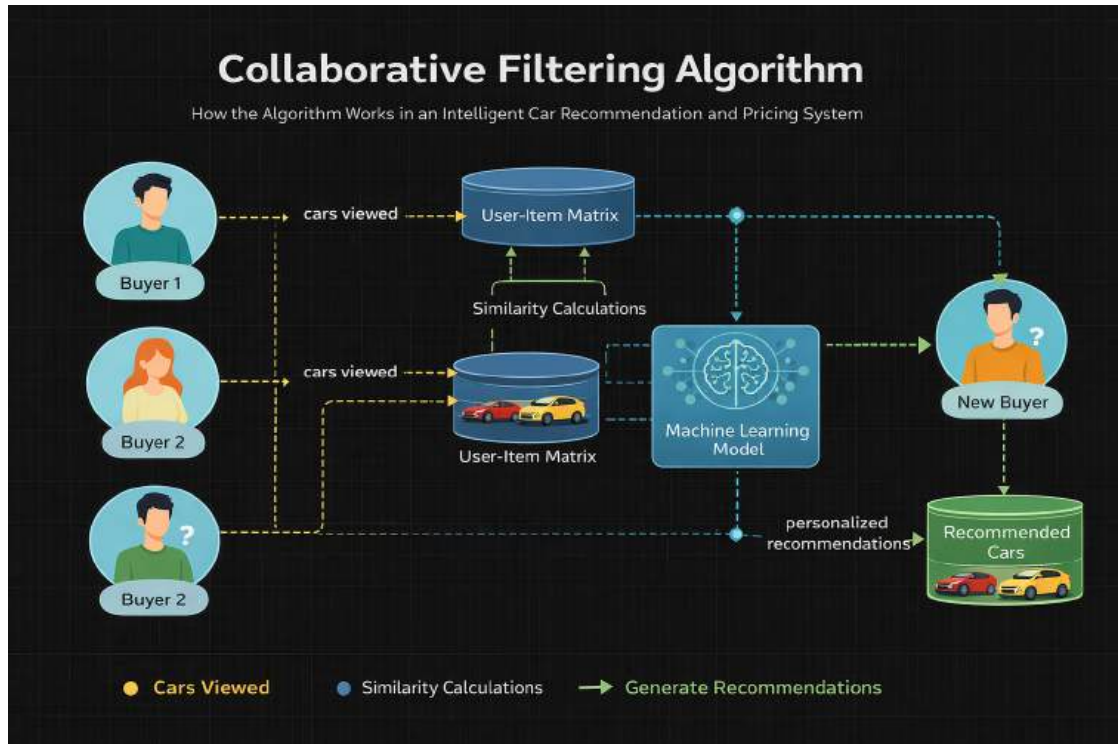


Figure 2.3: Collaborative Filtering Algorithm

2.2.4 Technological Methods in the Development of Online Car Marketplaces

Modern online car marketplaces use web technologies, databases and analytic models to handle vehicle listings along with user interactions. Traditional platforms emphasize primarily static listing and manual filtering, and this restricts decision support. The integration of machine learning techniques brings even more value to such platforms by allowing intelligent recommendations, as well as price prediction, improving the efficiency, and user experience of these platforms.

2.2.5 Machine Learning Techniques for Price Prediction

Machine learning-based regression techniques are common for predicting the price of cars through analysing the relationship between attributes of vehicles and the past pricing data. Attributes such as brand, year of manufacture, mileage as well as engine

type greatly influence the prices of cars. Regression models make price estimation by data more lenticular way and support transparency and informed decision making for buyers and sellers.

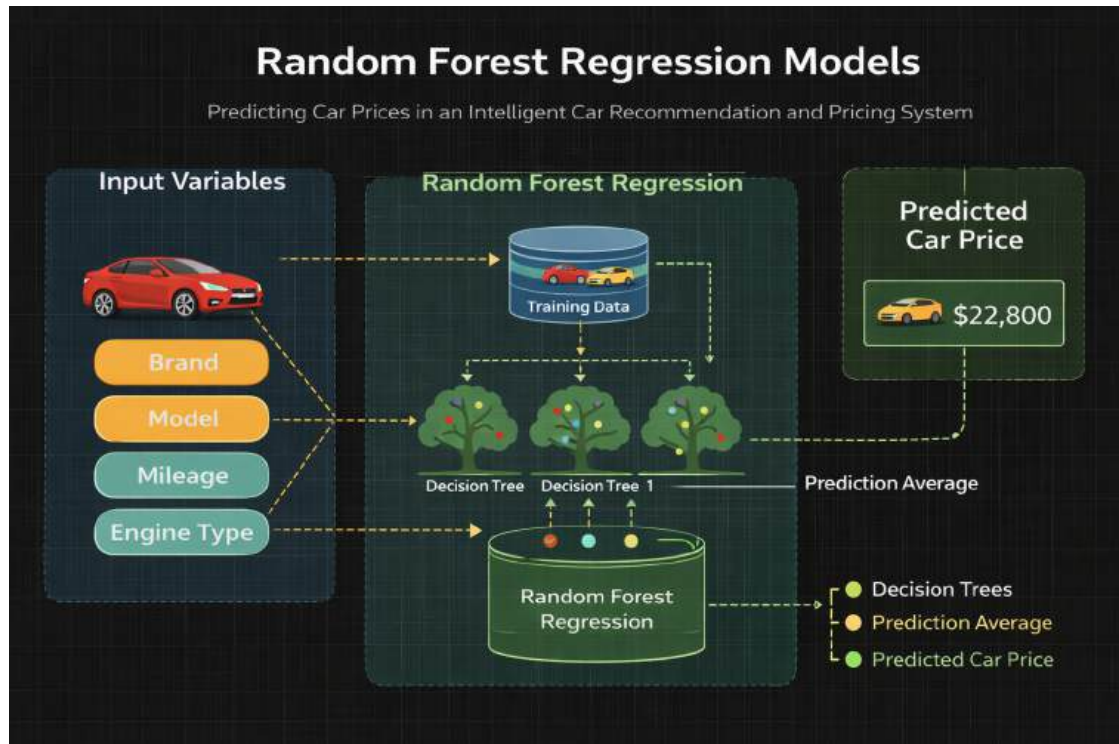


Figure 2.4:Random Forest Regression

2.2.6 Consumer Trust and Behaviour in Online Car Marketplaces

Consumer trust is key with online car transactions. Buyers are more likely to interact with platforms that have clear pricing and credible recommendations. Intelligent systems that provide individual advices and data-based pricing estimates can make users feel more confident and less uncertain when purchasing products.

2.3 Existing Online Car Sales

There are some existing online car sales platforms that offer basic vehicle listing and search functionality. While these systems enable users to filter vehicles based on certain criteria like brands and pricing range, the systems are not intelligent to suggest recommendations and set automated pricing efficiently. Most currently available

platforms rely on prices set by the sellers and manual browsing, which can explore inefficiencies that result in reduced user satisfaction. These limitations make the need of intelligent systems, combining recommendation and pricing systems.

2.4 Article Review

This section goes through some of the academic studies connected with car recommendation system and price prediction using machine learning. Previous research works have shown that collaborative filtering proves more effective in making recommendations more relevant than regression models that accurately estimate prices if trained on the right data sets. However, many studies examine recommendation or pricing separately instead of combining both into one whole. The articles studied confirm the feasibility of the application of machine learning techniques in automotive marketplaces, as well as opportunities for system integration.

Table 2.4: Literature Review

Title, Authors & Year	Aims / Objectives	Method / Technique	Advantages	Disadvantages
Price Prediction for Pre-Owned Cars Using Ensemble Machine Learning Techniques (Chetna Longania, Sai Prasad Potharaju, Sandhya Deore, 2021)	To develop a system that predicts fair prices for pre-owned cars in the Mumbai region using machine learning models.	Ensemble ML methods: Random Forest and XGBoost regression. Data collected from car websites and processed using	XGBoost achieved high accuracy (RMSE = 0.53). Handles missing values effectively. Random Forest prevents overfitting and provides robustness.	Dataset limited to Mumbai region, reducing generalisation. High computational cost. Performance depends on data quality.

		Python (Anaconda, sklearn, Flask).		
A Machine-Learned Ranking Algorithm for Dynamic and Personalised Car Pooling Services (Mattia Giovanni Campana et al., 2023)	To design a dynamic and personalised carpooling recommendation system that adapts to user behaviour over time.	Learning-to-Rank (LR), online learning, SGD. Used Twitter and Foursquare data. Implemented as an Android app.	Learns preferences dynamically. High prediction accuracy. Supports real-time personalisation.	Requires large interaction data. Computationally intensive. Evaluation limited to simulated data and pilot testing.
Car Price Prediction Using Machine Learning Techniques (Amneh J. Kenana et al., 2024)	To design a robust car price prediction model by comparing seven ML algorithms.	Random Forest, Linear Regression, KNN, AdaBoost, SVM, XGBoost, Decision Tree. Used Label Encoding, RFE, RMSE, R ² .	Random Forest achieved best accuracy. Large dataset (~46,000 records). Strong preprocessing pipeline.	Limited to structured datasets. High preprocessing effort. Some datasets contain missing and duplicate values.

Pre-Owned Car Price Prediction by Employing Machine Learning Techniques (Mauparna Nandan & Debolina Ghosh, 2023)	To forecast used car prices and determine the optimal ML algorithm.	Linear Regression, Lasso, Ridge, KNN, Random Forest, AdaBoost, XGBoost. Kaggle dataset (~426,880 records).	XGBoost achieved highest accuracy (86.87%). Effective preprocessing and feature engineering.	Dataset limited to U.S. market. Excludes macroeconomic features. Manual feature engineering may miss complex relationships.
Price Prediction and Classification of Used Vehicles Using Supervised Machine Learning (Lucija Bukvić et al., 2022)	To estimate used-vehicle prices and classify vehicles in the Croatian market.	Linear Regression, Random Forest, SVM. Data mining from online marketplace.	Linear Regression achieved 95% accuracy. Model adaptable to other markets. Demonstrates importance of brand and mileage.	Limited dataset size and geographic scope. Excludes external economic factors. Short data collection period.
How Much Is My Car Worth? A Methodology for Predicting Used Cars' Prices Using Random Forest (Naburn	To develop a user-friendly system for predicting used car prices using supervised	Random Forest Regression. Kaggle eBay Kleinanzeigen dataset (~370,000	High training accuracy (95.82%). Captures complex relationships. Robust	Limited feature diversity. Risk of slight overfitting. Dependent

Pal et al., 2018)	ML.	records). Feature importance analysis.	ensemble averaging.	on Kaggle dataset quality.
Predicting Used Car Price Based on Machine Learning (Jiayi Lin, 2024)	To identify the most effective regression model for used car price prediction.	Linear Regression, Ridge Regression, Random Forest. Data preprocessing and residual analysis.	Random Forest achieved highest accuracy. Strong interpretability. Visual evaluation improved reliability.	Small dataset from single source. Possible overfitting. Limited categorical feature handling.
Pre-Owned Car Price Prediction: A Machine Learning Approach (Thanu V. Marholkar et al., 2025)	To build a predictive model estimating resale prices of pre-owned cars.	Linear Regression, Ridge Regression, Lasso Regression. Kaggle dataset. Feature selection on brand, mileage, transmission.	Linear Regression achieved highest accuracy ($R^2 = 0.915$). Simple and interpretable.	No ensemble or deep learning models used. Limited generalisation. Misses nonlinear relationships.
Used Car Price Prediction Using	To predict the resale value of	Linear Regression,	Random Forest and	Dataset limited to

Machine Learning Techniques (S. Varma, A. Kumar, University of Texas, USA, 2020)	used cars in the U.S. market using machine learning models to assist buyers and sellers.	Random Forest, Gradient Boosting. Dataset collected from U.S. online car marketplace s. Evaluation using RMSE and R^2 metrics.	Gradient Boosting achieved high prediction accuracy. Handles nonlinear relationships well. Demonstrates effectiveness of ensemble methods.	U.S. market only. No integration with recommendation systems. Does not consider user behaviour or preferences.
Predicting Used Car Prices with Machine Learning Algorithms (M. Hutter, J. Müller, Technical University of Munich, Germany, 2021)	To analyse the effectiveness of different regression-based ML models for used car price prediction in the European market.	Decision Tree, Random Forest, Support Vector Regression (SVR). Dataset obtained from European car sales platforms. Feature engineering on mileage, age, fuel type.	Random Forest achieved best performance. Strong feature importance analysis. Applicable to European markets.	Limited dataset size. Market-specific results may not generalize globally. No external economic factors included.

2.5 Summary Of The Chapter

This chapter reviewed theories, techniques and existing studies related to recommendation systems, pricing prediction, and online automobile marketplace. The review uncovered significant limitations in current systems especially the lack of integrated system functionality for recommendations and pricing.

Based on the literature survey collaborative filtering and regression based machine learning models were found as the best machine learning techniques for the proposed system. The next chapter will describe the methodology, which will be used for the design and implementation of the Intelligent Car Recommendation and Pricing System Using Machine Learning.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the methodology used in the development of the Intelligent Car Recommendation and Pricing System using the Machine Learning. It explicates the lifecycle of the project, requirements analysis, system design, data models, proof-of-concept validation, and complexity of solutions. The methodology provides a systematic framework for guaranteeing that the system is designed in a systematic way that meets the requirements of the user and realises the objectives specified in Chapter 1.

3.2 Project Life Cycle

The Waterfall Model has been adopted as the project life cycle in this investigation. The sequential approach to development is comprised of simply defined phases and each phase needs to be completely accomplished before moving to the next phase. The phases include the requirement analysis, system design, implementation, testing, deployment, and maintenance phase. The model is considered to be appropriate for the present undertaking as the system requirements are unambiguously defined at the beginning and the project scope is under strict control. Moreover, the Waterfall approach to the project makes it easier to document and evaluate at every stage of the development process.

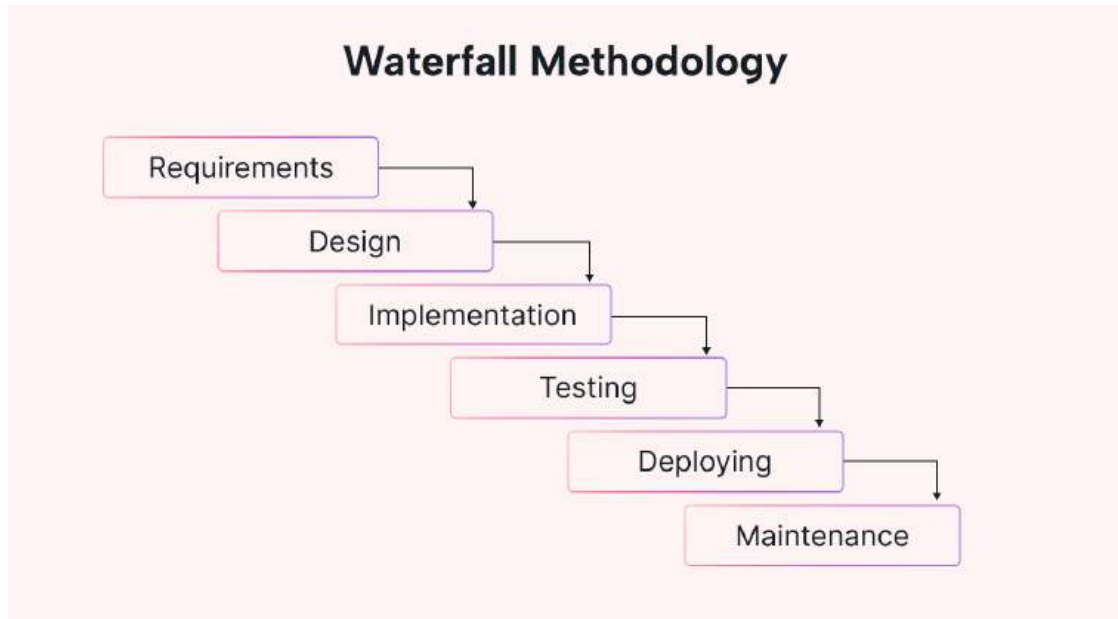


Figure 3.1: Waterfall Methodology

3.3 Requirements Analysis

This part outlines the functional and non-functional needs of the Intelligent Car Recommendation and Pricing System using Machine Learning. The requirements are based on the system objectives and user requirements.

3.3.1 Functional Requirements

Table 3.1: Functional Requirements

No.	Requirement Name	Description
FR1	User Registration	The system makes it easy for buyers and sellers to sign up for the system through the provision of verifiable personal data.
FR2	User Login	The system allows the

		registered users to authenticate and access the platform by using their verified credentials.
FR3	Profile Management	Users can view and update their personal profile information.
FR4	Car Listing Management	Sellers can add, update, and remove car listings with detailed vehicle attributes.
FR5	Car Search	Buyers can browse and search available car listings based on selected criteria.
FR6	Car Recommendation	The system generates personalised car recommendations using collaborative filtering.
FR7	Price Prediction	The system predicts car prices using machine learning regression models.
FR8	View Predicted Prices	Buyers and sellers can view predicted car prices for comparison and decision support.
FR9	User Management	Administrators can manage buyer and seller

		accounts.
FR10	System Monitoring	Administrators can monitor car listings and system data for maintenance purposes.

3.3.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

No.	Requirement Type	Description
NFR1	Usability	The system should be easy to use with a clear and intuitive user interface.
NFR2	Performance	The system should generate recommendations and price predictions within an acceptable response time.
NFR3	Security	User data must be securely stored and protected from unauthorized access.
NFR4	Reliability	The system should function consistently without unexpected failures.
NFR5	Data Accuracy	The system should ensure

		the accuracy and integrity of stored data.
NFR6	Scalability	The system should support future expansion in terms of users and data volume.

3.4 Software and Hardware Requirements

This section solves the software and hardware requirements that are required for the development, testing, and implementation of the Intelligent Car Recommendation and Pricing System Using Machine Learning. The requirements specified ensure that the system is working efficiently and helps to implement machine learning models.

3.4.1 Software Requirements

The software tools and technologies needed for this project are listed in Table 3.3.

Table 3.3: Software Requirements

No.	Software	Description
1	Operating System	Windows 11
2	Programming Language	Python (for machine learning and backend development)
3	Web Framework	Flask
4	Front-End Technologies	HTML, CSS, JavaScript
5	Database Management System	MySQL
6	Machine Learning Library	Scikit-learn

7	Code Editor / IDE	Visual Studio Code
8	Web Browser	Google Chrome
9	Version Control	Git (optional)

3.4.2 Hardware Requirements

The minimum hardware specifications required for system development and execution are shown in Table 3.4.

Table 3.4: Hardware Requirements

No.	Hardware	Specification
1	PC	• Processor: AMD Ryzen 5 7500F • RAM: 16 GB • Storage : 512 GB • GPU : RTX4060 • OS : Windows 11
2	Laptop (Asus Rog Strix G15)	• Processor: AMD Ryzen 7 4800H • RAM: 16 GB • Storage : 1 TB • GPU : RTX3050 • OS : Windows 11

3.5 System Design

The following section describes the general structure of Intelligent Car Recommendation and Pricing System based on Machine Learning methods. This design phase focuses on the specification of the system architecture, data flow and data structure to ensure that the system can operate with optimum efficiency and meet the stipulated requirements. The design is presented in the form of framework diagram, Data Flow Diagrams (DFD) and Entity Relationship Diagram (ERD).

3.5.1 Framework

The framework represents the exchanges between the users and the central components of the system. The system is made up of three main user roles, Buyer, Seller and Administrator. Each of the roles uses a web-based interface to access the system. The Buyer module offers the functionality to enable users to register, authenticate, find available car listings, get tailored car recommendations as well as view predicted available car prices. The Seller module allows sellers to integrate car listings as well as submit requests for prices that are predicted based on attributes of the vehicle. The Administrator module has the job of overseeing user accounts and monitoring car listings as well as ensuring the validity of data.

At the heart of the system is the processing layer that coordinates data validation, recommendation production and price prediction. This core resides in interaction with the database to perform persistence and retrieval of user information, vehicle data and pricing data. Additionally, with the availability of cars and comparative market data, the system may have access to external data sources.

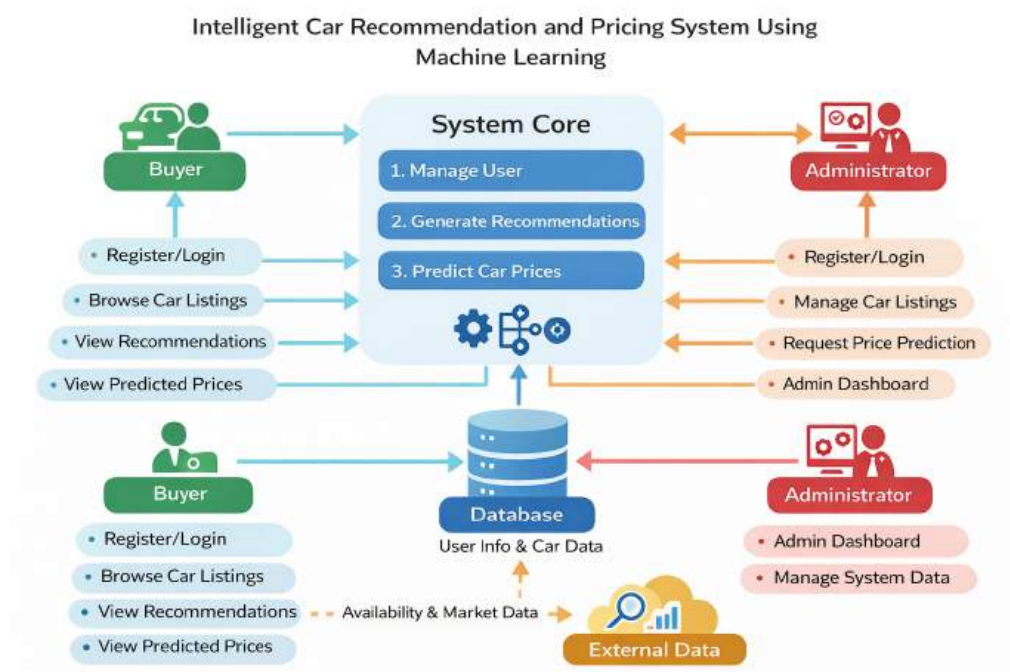


Figure 3.2.1 Framework

3.5.2 Process Model

The Data Flow Diagram (DFD) displays the flow of data within the system as well as the interactions between processes, external entities, and the data stores.

3.5.2.1 Context Diagram

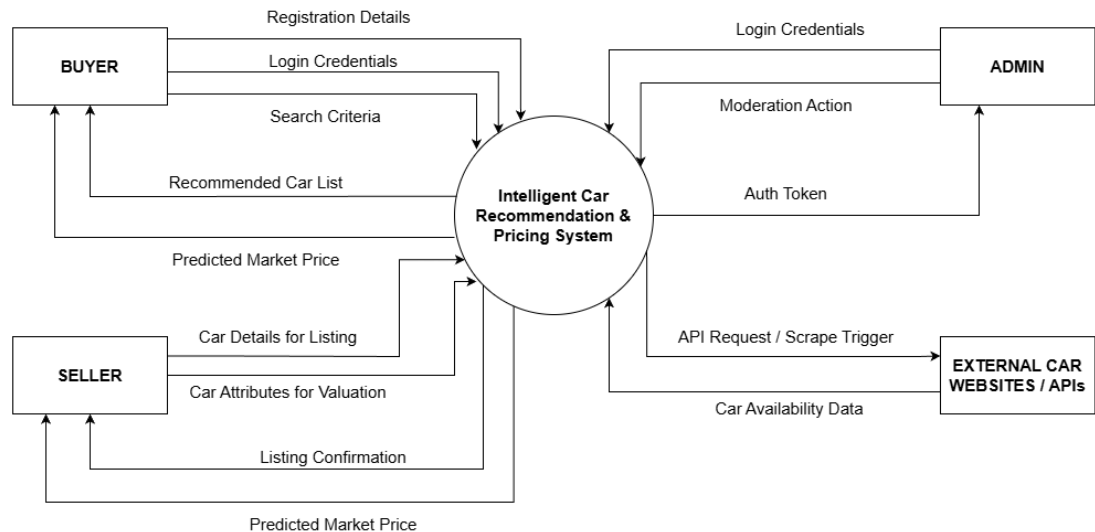


Figure 3.2.2 Context Diagram

The Context Diagram provides a macroscopic view of the Intelligent Car Recommendation and Pricing System by describing it as a single process. It shows the interactions between the system and external entities, i.e. the Buyer, Seller, Administrator and the external entities of the automotive websites or APIs. Buyers fill in login properties and search requests to the system and receive the recommended datasets on cars and predicted prices as outputs. Sellers enter the vehicle information and receive proposed price predictions from the system. Administrators give management commands and get system reports. Externally related data forms like car availability or market data for supporting the functionality of the system.

3.5.2.2 Data Flow Diagram (DFD) Level 0

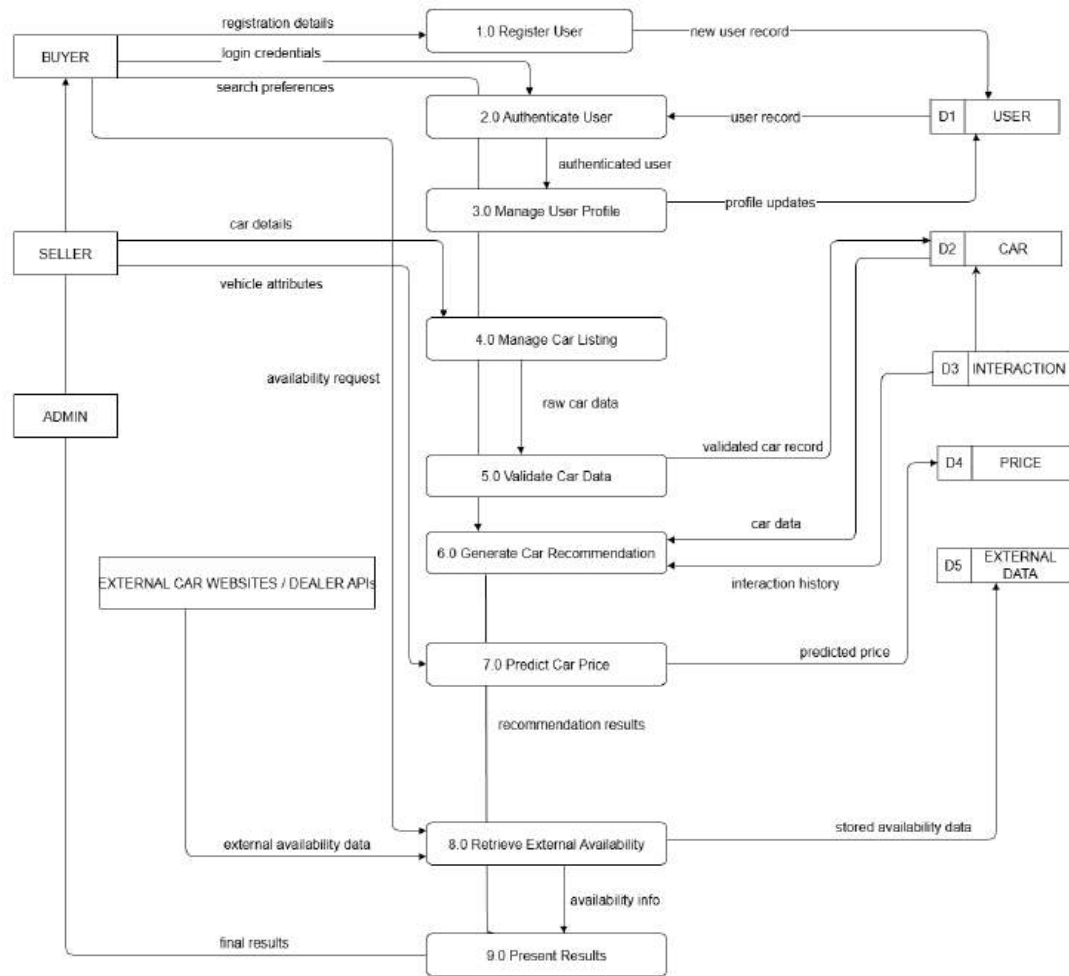


Figure 3.2.3 DFD Level 0

The DFD Level 0 is used to expand the system, by showing the main data flows to and from external entities, in the form of a single process. It focuses on a key input/output data exchanged between the system and each external entity. Buyers interact with the system by submitting login information and questions, and they receive recommended car listings and expected prices. Sellers provide vehicle details and obtain proposed pricing details. Administrators communicate through management commands and receive data for monitoring from the system. External data sources are used to provide external information about markets or availability for improving processing.

3.5.2.3 Data Flow Diagram (DFD) Level 1

Data Flow Diagram Level 1 Process 4.0 is Seller Car Listing Management in Intelligent Car Recommendation and Pricing System using Machine learning. The process starts when the Seller provides the vehicle information - including pictures, condition and asking price - to the system. In Process 4.1 (Validate Seller and Car Information), the system authenticates the seller's identity from the data in the USER (D1) data store, and it authenticates the vehicle information submitted for the process to make sure that it's complete and accurate.

Upon successful validation, the system moves on to Process 4.2 (Create New Car Listing) in which a new vehicle listing record is created to be stored in the CAR (D2) data store. The system then returns a listing confirmation, or an error message, to the seller. Sellers may also request to see existing listings which is handled in Process 4.3 (Retrieve Existing Listings) by retrieving relevant vehicle records from the CAR (D2) data store and displaying them.

Additionally, around sellers can update or remove existing listings by way of Process 4.4 (Update or Remove Car Listing). In this stage, the system updates or deletes the selection of vehicle from the CAR (D2) data store and returns the updated status to the seller. This structured workflow ensures the secure verification of sellers, the accurate storage of the data and the consistent management of vehicle listings.

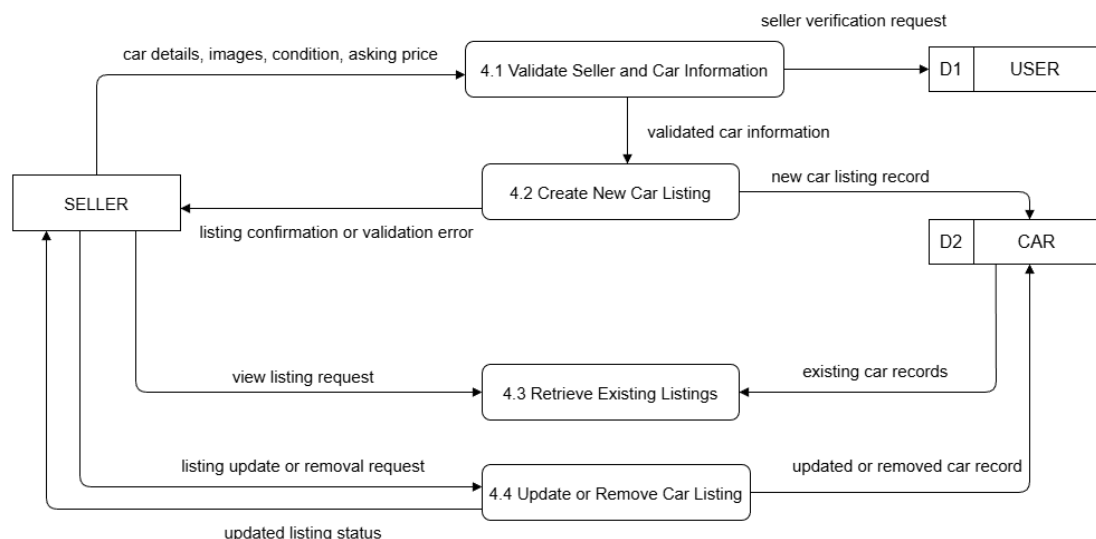


Figure 3.3.4 DFD 1 Process 4.0

Data Flow Diagram Level 1 Process 5.0 The Buyer Car Recommendation

Procedure in the Intelligent Car Recommendation and Pricing System using Machine Learning. The procedure begins when the Buyer has an interaction with the system by feeding preferences and browsing behaviour (previously viewed cars, saved listings, search activities etc.). In Process 5.1 (Capture Preferences and Behaviour), this information is recorded in the data store called the INTERACTION (D3) data store in the system.

After this, the system moves to Process 5.2 (Generate Recommendation Ranking). Here, the system uses the data stores for the history of interactions (D3) and the vehicle information (D2) to access the information needed to retrieve the history of interactions and the vehicle information. Using these data, the recommendation engine uses machine learning techniques to analyse user behaviour and the attributes of the vehicles and hence arrives at a list of cars which are suited for the buyer in a ranked list.

Finally, Process 5.3 (Display Recommended Cars) displays the results of the ranked recommendation results to the buyer in the form of recommended vehicle listings. This flow ensures that the recommendations are personalized, data driven and continuously refined based on user interactions so as to enhance the buyer decision-making experience.

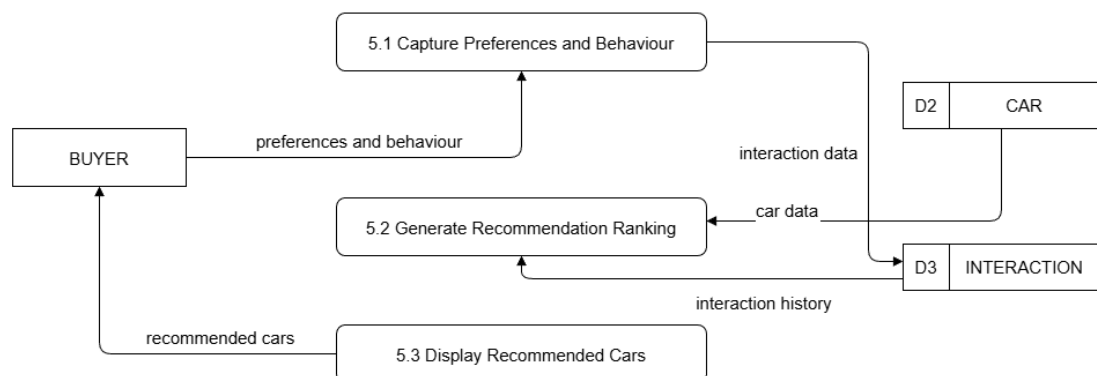


Figure 3.3.5 DFD 1 Process 5.0

Data Flow Diagram Level 1 Process 7.0 is the procedure of predicting and recommending price in the Intelligent Car Recommendation and Pricing System using Machine Learning. The process starts with the seller providing the vehicle attributes and wanted price for the vehicle to the system. In Process 7.1 (Collect Vehicle Attributes and Asking Price), the system collects relevant attributes of the vehicles, for example, brand, model, year, mileage and condition, and creates a feature dataset for the purpose of attitude prediction.

Next, in Process 7.2 (Predict Fair Market Price), the system uses a machine learning regression model based on the dataset of features to obtain a predicted fair market price. The predicted price is being stored as a price record in the PRICE data store. The system then goes on to Process 7.3 (Compare with Market Prices), in which the predicted price is compared with historical prices that are retrieved from the PRICE data store and current market prices that are retrieved from the EXTERNAL DATA data store. This comparison will provide pricing information and decision support by telling the seller whether the asking price of the items or services is competitive in the market.

Finally, Process 7.4 (Present Price Recommendation) brings together the results of the prediction and comparison and builds a clear price recommendation to the seller including proposed pricing suggestions based on machine learning predictions and market comparisons. The overall flow is ensured so that sellers receive data-driven and reliable pricing support to help them set competitive and fair vehicle prices.

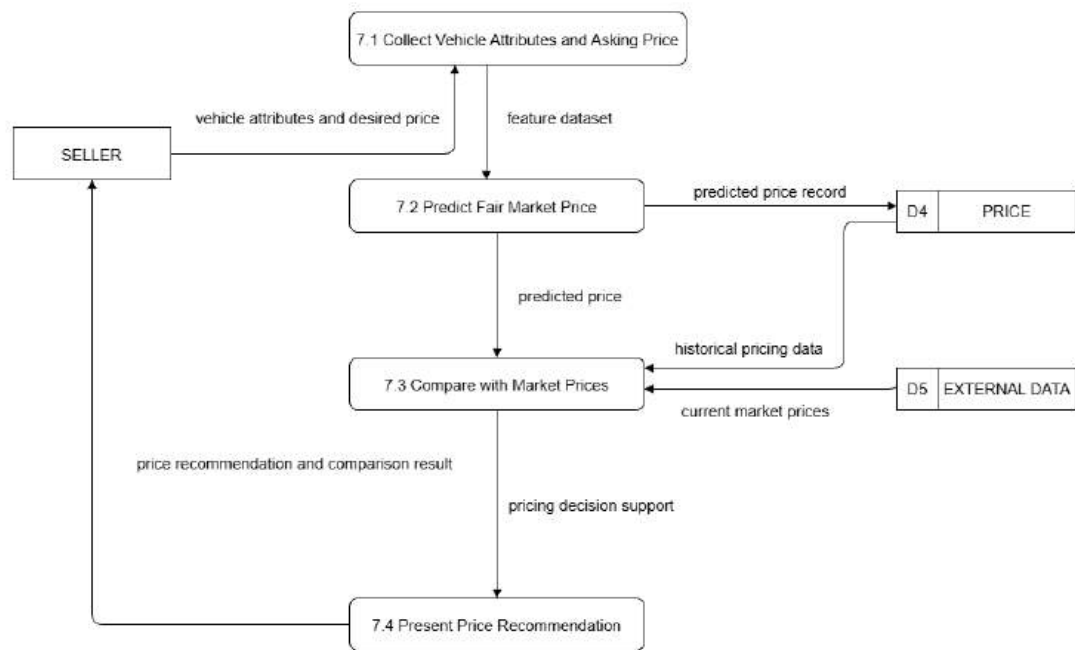


Figure 3.3.6 DFD Level 1 7.0

3.5.2.4 ENTITY RELATIONSHIP DIAGRAM (ERD)

The ERD is the representation of the database structure of Intelligent Car Recommend and Pricing System using Machine Learning to determine how main entities are related with each other to facilitate functioning of the system.

The USER entity is storing information for all the system users - i.e. buyers, sellers, and the administrators. Each user has a unique identification known as user ID and has attributes like name, email, password and roles. A user can list more than one vehicle and hence creates one to many relationship between USER and CAR. This relationship represents sellers that are listing vehicles in system. Additionally, a user can create reports and interact with vehicles, which creates one-to-many relationships between USER and REPORT, and between USER and INTERACTION.

The CAR entity represents the list of vehicles in the system and contains attributes such as brand, model, year, mileage, price and status. Each vehicle is listed by one user, but may match more than one record in other entities. A vehicle may have more than one price prediction, which is represented by a one-to-many relationship with the

PRICE_PREDICTION entity. This allows the system to maintain predicted prices that are generated by machine learning models and confidence values.

The INTERACTION entity captures the buyer's interactions with vehicle listings such as viewing, saving, or comparing. Each interaction record consists of an interaction id, type of interaction and time-stamp, connecting a particular user to a particular vehicle. This interaction data is vital to analyse the user behaviour and generate personalised recommendations.

The MARKET_PRICE entity is used to store external or historical market pricing information including average market prices. Each of the vehicles can be compared against multiple market price records, meeting price validation and current market trend comparison standards. This entity is used to integrate data outside the platform for more accurate pricing insights.

Finally, the REPORT entity is used for system generated reports related to usage, pricing or performance. Each report is created by a user, usually an administrator, and is distinguished by a report ID and report type.

Overall, the ERD helps in having the right data organisation and relationship between users, vehicles, interactions, price prediction, and market data. It supports key functions in the system e.g. vehicle listing management, personalised recommendations, machine learning-based prediction of price and system monitoring; while preserving data integrity and consistency.

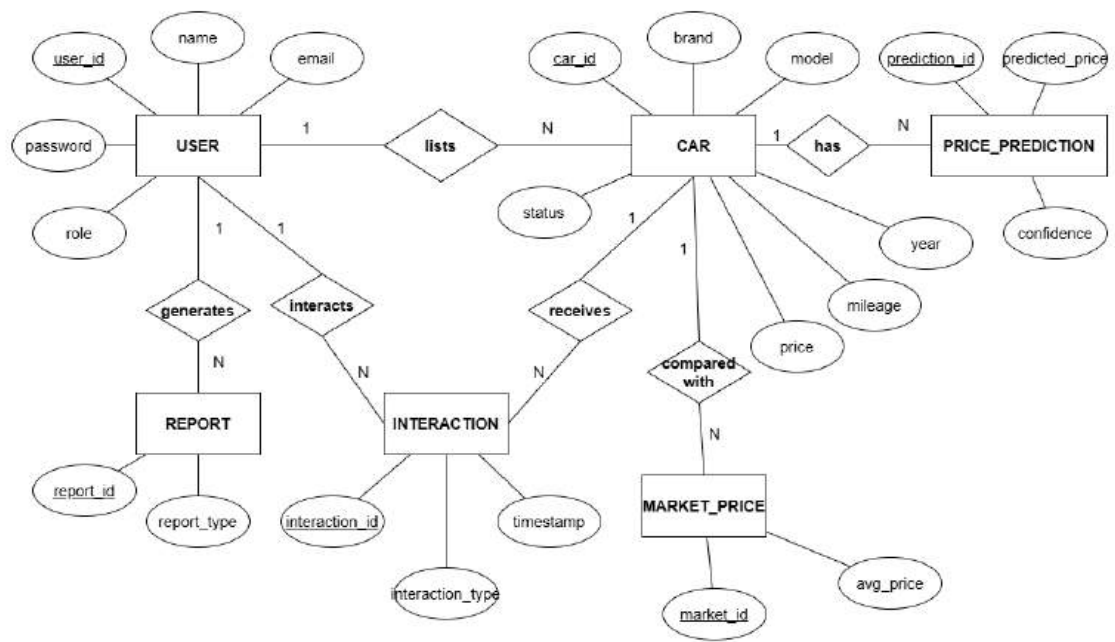


Figure 3.2.7 ENTITY RELATIONSHIP DIAGRAM (ERD)

3.5.2.5 Proof Of Concept

Figure 3.6.1 Registration Page

Registration and Login Flow

Flow 1: User Registration

1. The user accesses the system homepage.
2. The user selects the Register option.
3. The system displays a registration form requesting user details such as email, password, full name, and user type (Buyer or Seller).
4. The user submits the form.
5. The system validates the input and creates a new account.
6. The user is redirected to the login page upon successful registration.

Outcome:

A new user account is created successfully and stored in the system database.

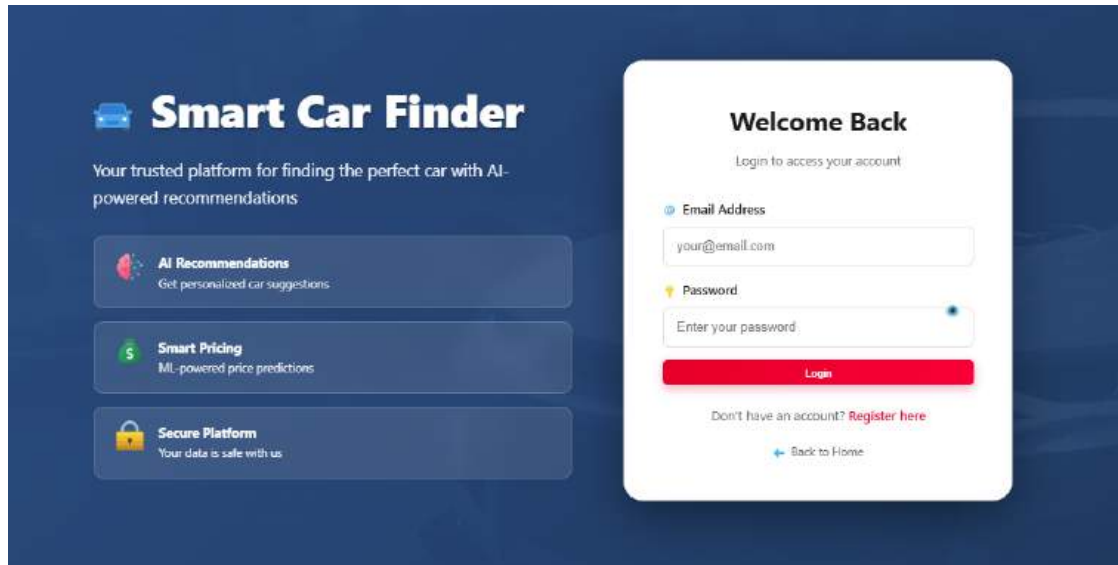


Figure 3.3.2 Login Page

Flow 2: User Login

1. The user navigates to the login page.
2. The user enters registered email and password.
3. The system authenticates the credentials.
4. Upon successful authentication, the user is redirected to the appropriate dashboard based on user role.

Outcome:

The user gains secure access to the system dashboard.

Buyer User Flows

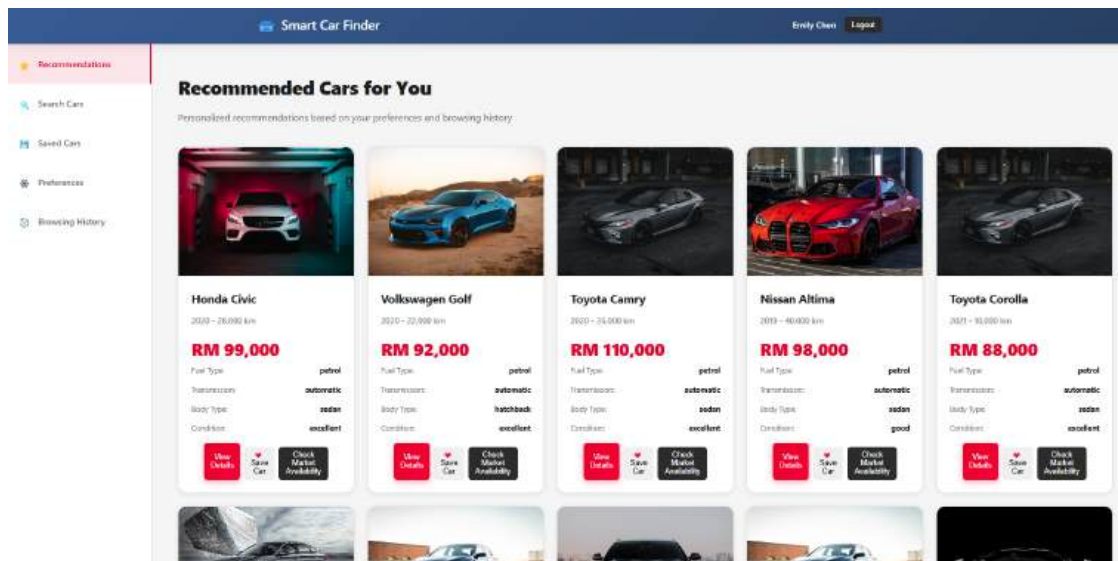


Figure 3.3.3 Recommendation Page

Flow 3: View Personalised Recommendations

1. The buyer logs into the system.
2. The buyer dashboard loads automatically.
3. The system retrieves user preferences and browsing history.
4. The recommendation engine generates a ranked list of cars using machine learning.
5. Recommended cars are displayed on the dashboard.

Outcome:

The buyer receives personalised car recommendations based on preferences and interaction history.

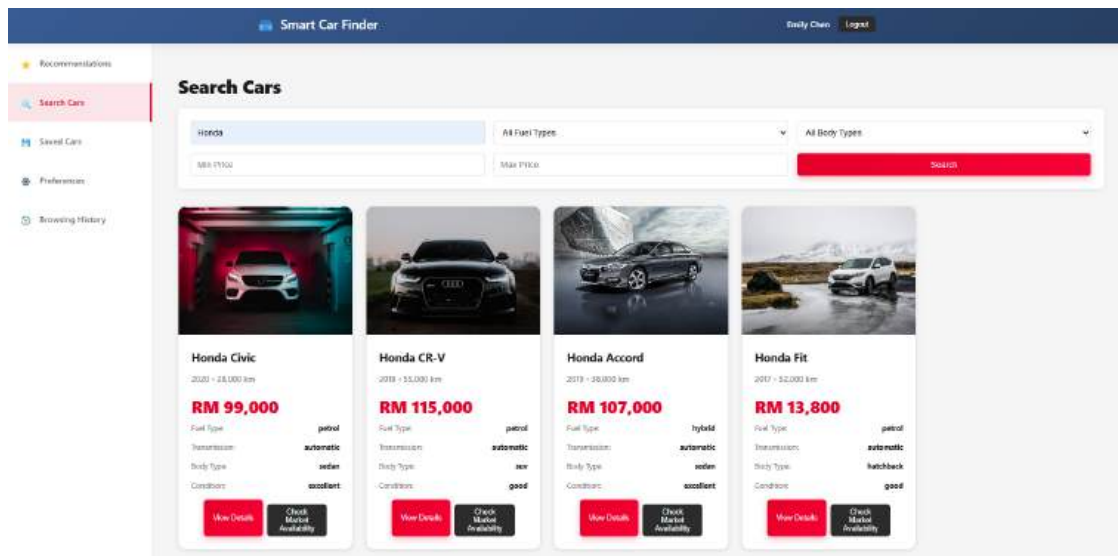


Figure 3.3.4 Search Car Page

Flow 4: Browse and Search Cars

1. The buyer selects the *Search Cars* option.
2. The system displays all available car listings.
3. The buyer applies filters such as brand, price range, fuel type, and body type.
4. The system updates the results dynamically based on selected filters.

Outcome:

Filtered car listings are displayed in real time.

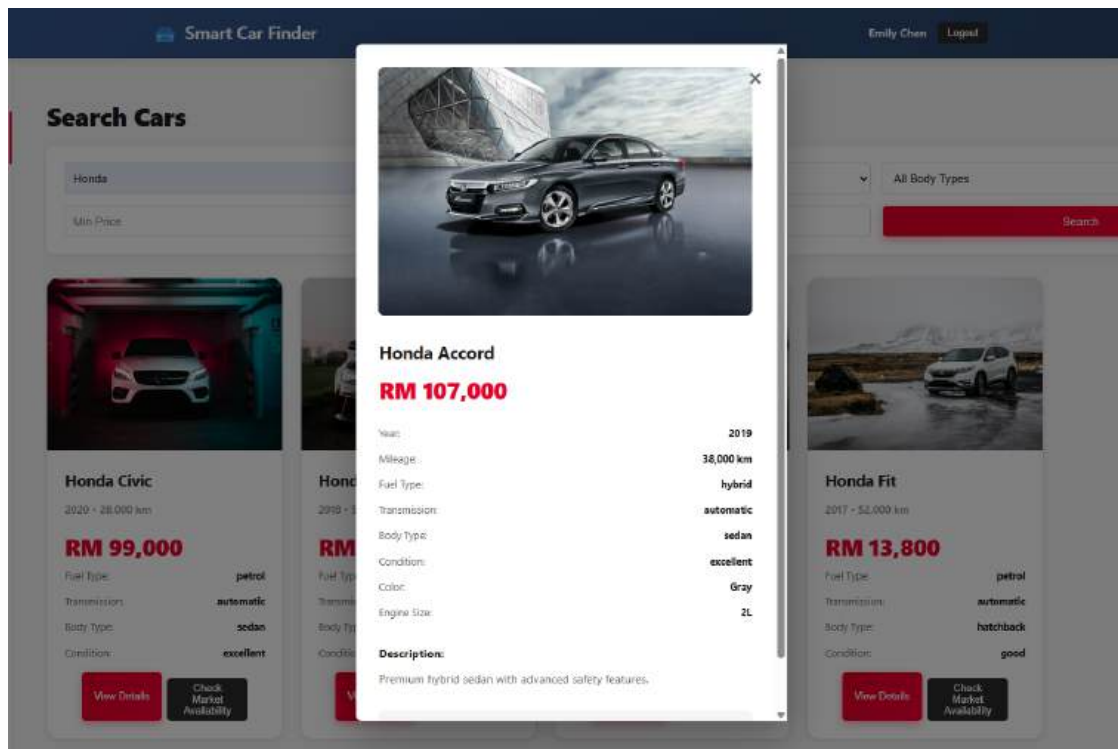


Figure 3.3.5 View Car Page

Flow 5: View Car Details

1. The buyer selects a car from the listing.
2. The system opens a detailed car information view.
3. Full specifications, pricing, and seller details are displayed.
4. The system records the view as browsing history.

Outcome:

The buyer can review complete vehicle information before making a decision.

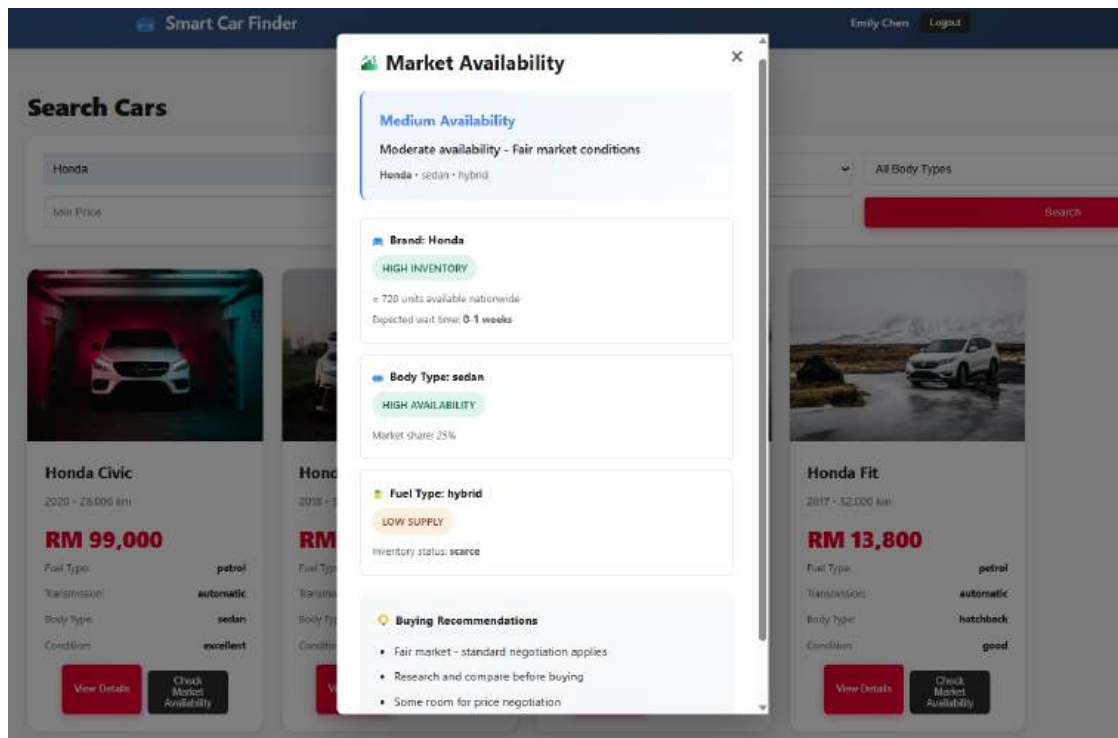


Figure 3.3.6 Check Market Availability Page

Flow 6: Check Market Availability

1. The buyer selects *Check Market Availability*.
2. The system retrieves external market reference data.
3. Market insights such as availability level, average price, demand level, and price comparison are displayed.

Outcome:

The buyer receives market-based insights to support informed decision-making.

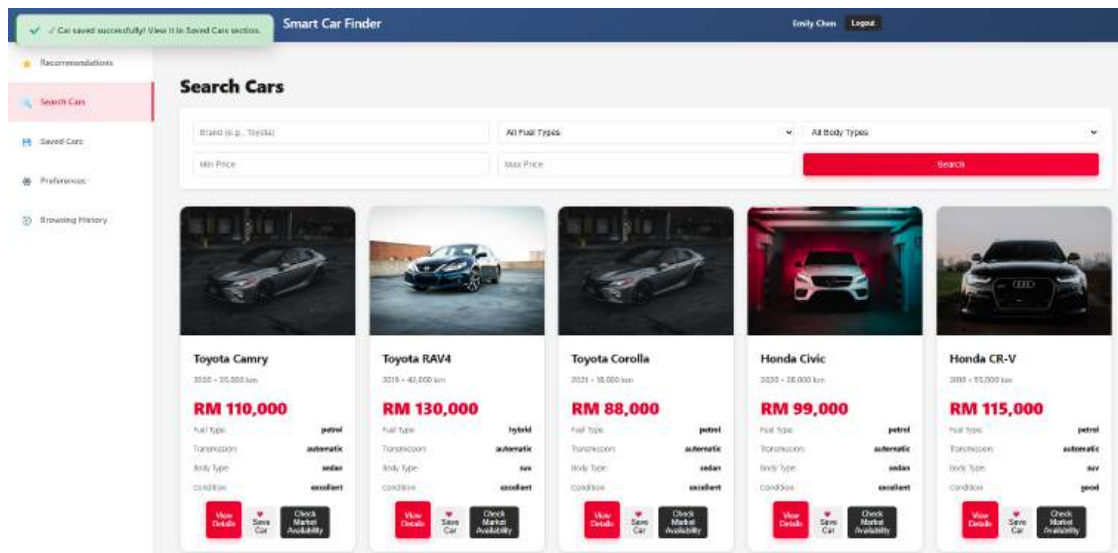


Figure 3.3.7 Save Car Page

Flow 7: Save Favourite Cars

1. The buyer selects the save (favourite) option on a car listing.
2. The system stores the car in the buyer's saved list.
3. A confirmation notification is displayed.

Outcome:

Favourite cars are saved for future reference.

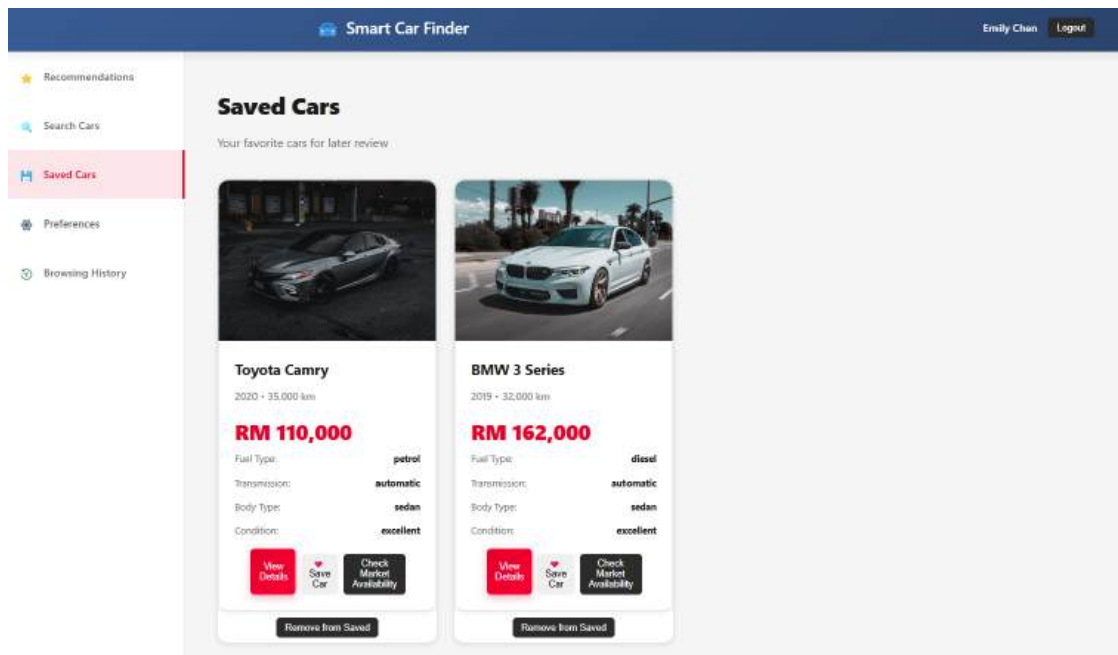


Figure 3.3.8 View Save Car Page

Flow 8: View Saved Cars

1. The buyer accesses the *Saved Cars* section.
2. The system retrieves and displays all saved cars.
3. The buyer can view details, remove favourites, or compare cars.

Outcome:

The buyer can manage shortlisted vehicles efficiently.

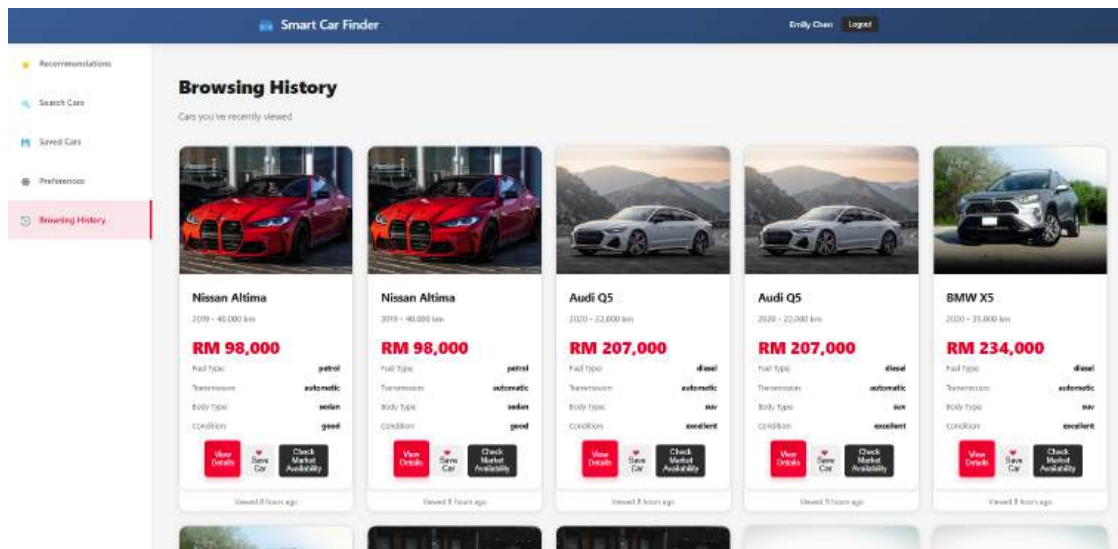


Figure 3.3.9 View Browsing Page

Flow 9: View Browsing History

1. The buyer selects the *Browsing History* option.
2. The system displays a chronological list of previously viewed cars.
3. The buyer can re-access car details from the history list.

Outcome:

Browsing history supports user recall and improves recommendation accuracy.

Figure 3.3.10 Preferences Page

Flow 10: Update Preferences

1. The buyer accesses the *Settings* page.
2. The buyer updates preferences such as budget range, brand, fuel type, and body type.
3. The system saves the updated preferences.
4. Recommendations are refreshed automatically.

Outcome:

The system adapts future recommendations based on updated preferences.

3. Seller User Flows

Smart Car Finder

My Listings

Add New Car

Price Checker

Market Insights

Add New Car Listing

List your car and get AI-powered price suggestions!

Brand *
Hyundai

Year *
2020

Mileage (km) *
9000

Fuel Type *
Petrol

Body Type *
Hatchback

Color
Black

Description
Describe your car's features, condition, and any additional information...

Model *
i30

Year Price (\$) *
19387.62

Engine Size (L)
1.5

Transmission *
Automatic

Condition *
Excellent

AI Predicted Fair Price: RM 19,308
This is the estimated fair market value based on your car's specifications.

Get Price Prediction List Car

Figure 3.3.11 Add Car Page

Flow 11: Add New Car Listing

1. The seller logs into the seller dashboard.

2. The seller selects *Add New Car*.
3. The seller enters car details such as brand, model, year, mileage, and condition.
4. The seller requests a price prediction.
5. The system generates a predicted price using a machine learning model.
6. The seller submits the listing.

Outcome:

A new car listing is published with data-driven pricing support.

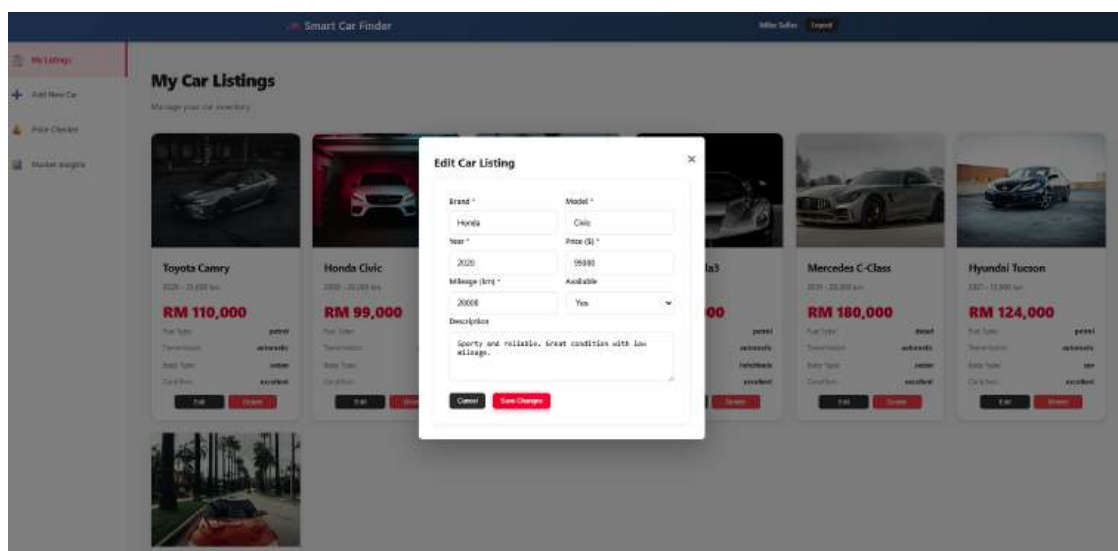


Figure 3.3.12 Edit or Delete CarPage

Flow 12: Edit or Delete Listing

1. The seller selects an existing listing.
2. The seller edits details or chooses to delete the listing.
3. The system updates or removes the listing accordingly.

Outcome:

Sellers can manage and maintain their listings effectively.

4. Administrator User Flows

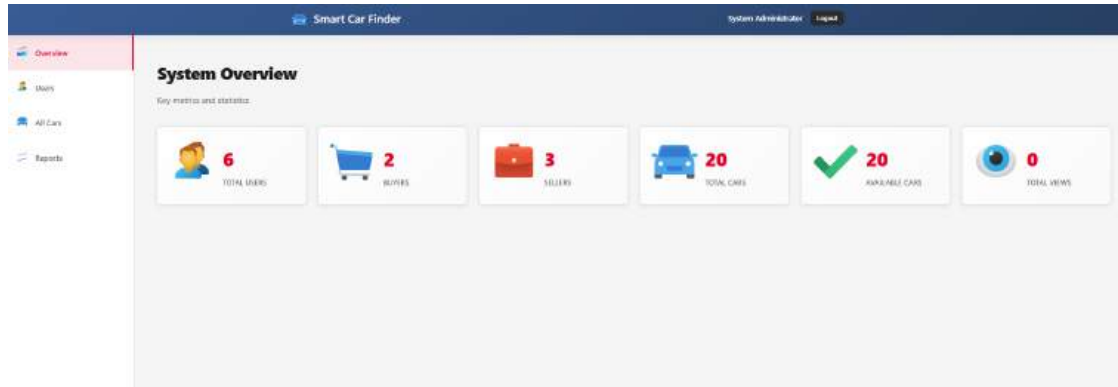


Figure 3.3.13 Admin Dashboard

The screenshot displays the 'User Management' section of the 'Smart Car Finder' admin dashboard. The dashboard has a dark blue header with the site name and user role ('System Administrator'). A left sidebar contains navigation links: Overview, Users (selected), All Cars, and Reports. The main content area shows a table of registered users with management actions.

ID	Name	Email	Role	Created At	Actions
1	System Administrator	admin@systemupgrd.com	ADMIN	18 Jun 2025	Profile
2	Emily Chen	buyer1@stsc.com	BUYER	18 Jun 2025	Details
3	David Martinez	buyer2@stsc.com	BUYER	18 Jun 2025	Details
4	Mike Selzer	seller@stsc.com	SELLER	18 Jun 2025	Details
5	Premium Auto Dealer	dealer1@example.com	SELLER	18 Jun 2025	Details
6	City Car Sales	dealer2@example.com	SELLER	18 Jun 2025	Details

Figure 3.3.14 User Management Page

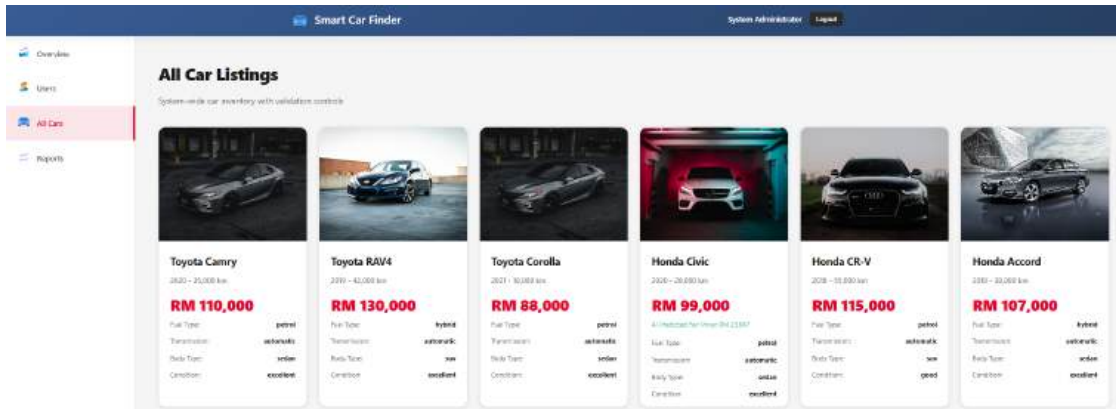


Figure 3.3.15 Car Listing Page



Figure 3.3.16 System Report Page

Flow 13: System Management

1. The administrator logs into the admin dashboard.
2. The system displays system statistics such as total users and total listings.
3. The administrator manages users and moderates car listings.

4. The administrator monitors system and model performance.

Outcome:

System integrity and data quality are maintained.

3.5.3 Data Dictionary

3.5.3.1 Table: USER

Table 3.5.2: Table USER

Field Name	Data Type	Length	Key Type	Description
user_id	INT	11	Primary Key	Unique identifier for each user
name	VARCHAR	100	-	Full name of the user
email	VARCHAR	100	Unique	User's registered email address
password	VARCHAR	255	-	Encrypted password for authentication
role	VARCHAR	20	-	User role (Buyer, Seller, Admin)

3.5.3.2 Table: CAR

Table 3.5.2: Table CAR

Field Name	Data Type	Length	Key Type	Description
car_id	INT	11	Primary Key	Unique identifier for each car
brand	VARCHAR	50	—	Car manufacturer brand
model	VARCHAR	50	—	Car model name
year	INT	4	—	Manufacturing year of the car
mileage	INT	10	—	Total distance travelled by the car
price	DECIMAL	10,2	—	Listed price of the car
status	VARCHAR	20	—	Availability status (Available, Sold, Hidden)
user_id	INT	11	Foreign Key	References USER(user_id)

3.5.3.3 Table: PRICE_PREDICTION

Table 3.5.3: Table PRICE_PREDICTION

Field Name	Data Type	Length	Key Type	Description
prediction_id	INT	11	Primary Key	Unique identifier for price prediction

predicted_price	DECIMAL	10,2	—	Machine learning predicted car price
confidence	DECIMAL	5,2	—	Confidence level of the prediction
car_id	INT	11	Foreign Key	References CAR(car_id)

3.5.3.4 Table: INTERACTION

Table 3.5.4: Table INTERACTION

Field Name	Data Type	Length	Key Type	Description
interaction_id	INT	11	Primary Key	Unique identifier for user interaction
interaction_type	VARCHAR	50	—	Type of interaction (View, Save, Compare)
timestamp	DATETIME	—	—	Date and time of the interaction
user_id	INT	11	Foreign Key	References USER(user_id)

car_id	INT	11	Foreign Key	References CAR(car_id)
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3.5.3.5 Table: MARKET_PRICE

Table 3.5.5: Table MARKET_PRICE

Field Name	Data Type	Length	Key Type	Description
market_id	INT	11	Primary Key	Unique identifier for market price record
avg_price	DECIMAL	10,2	—	Average market price from external sources
car_id	INT	11	Foreign Key	References CAR(car_id)

3.5.3.6 Table: REPORT

Table 3.5.6: Table REPORT

Field Name	Data Type	Length	Key Type	Description
report_id	INT	11	Primary Key	Unique identifier for system report

report_type	VARCHAR	50	—	Type of report (Usage, Pricing, System)
user_id	INT	11	Foreign Key	References USER(user_id)

3.4 System Complexity

The Intelligent Car Recommendation and Pricing System Utilising Machine Learning represents a moderate level of system complexity due to the deployment of diverse functional units, sophisticated machine learning algorithms, and different sources of heterogeneous data in a unified web-based system.

Firstly, the complexity of the system lies in the implementation of the machine learning-based recommendation mechanisms. The recommendation module uses collaborative filtering, content-based filtering, and hybrid filtering techniques to provide personalised recommendations for automobiles. This requires the capture of user preferences, maintenance of interaction histories, computation of similarity measures and dynamic updating of recommendation rankings. Compared to conventional rule-based or keyword-based search systems, the complexity of these operations greatly increases in both computing resources and design work.

Secondly, the price prediction module contributes to further system complexity by the implementation of the Random Forest regression models. The system needs to perform data preprocessing, feature extraction, model training, and predictions based on the attributes of the vehicles used, such as brand, model, year, mileage, and engine type. A further complication of overall system complexity is maintaining model fidelity while concurrently ensuring acceptable response time.

Thirdly, the system includes the ability to import market data from external sources such as third-party automobile websites or APIs to enable market availability checks and price checks. This integration requires the addition of extra data-retrieval,

data-preprocessing and data-validation procedures to ensure consistency between system data and external reference data, thus adding some further levels of complexity.

From an architectural perspective, the system allows for a variety of user roles, such as buyers, sellers and administrators, with separate interaction flows and access rights. Managing different user permissions, customized dashboards, and varying data interactions creates higher design complexity when compared to single user systems.

Despite these complexities, the system is tractable because the scope of capabilities is constrained, which is restricted to decision support functionalities. The lack of payment processing, transactional management, or vehicle ownership transfer mechanism is to limit technical and security complexity.

CHAPTER 4

IMPLEMENTATION AND RESULT

4.1 Introduction

This chapter presents the implementation and testing of the **Intelligent Car Recommendation and Pricing System Using Machine Learning**. The implementation phase focuses on transforming the system design into a fully functional web-based application, while the testing phase ensures that all system functions operate correctly according to the project requirements. The developed system aims to assist buyers in finding suitable vehicles through personalised recommendations and help sellers estimate competitive car prices using machine learning techniques.

The system is developed using the Laravel framework with MySQL as the database management system. Bootstrap is used to create a responsive and user-friendly interface that allows users to access the system from different devices. The system consists of three main modules: Buyer, Seller, and Administrator. Each module provides different functionalities based on the user's role, including managing car listings, browsing vehicles, generating price predictions, receiving personalised recommendations, and managing user accounts.

The intelligent features of the system are powered by collaborative filtering for car recommendations and Random Forest Regression for car price prediction. These machine learning models analyse user interactions and vehicle information to provide relevant recommendations and estimated market prices. The following sections present the interface design and testing results to demonstrate the functionality and performance of the developed system.

4.2 Interface Design

This section presents the user interface design of the Intelligent Car Recommendation and Pricing System. The front-end interface is constructed utilizing the Bootstrap grid system and components, establishing a consistent aesthetic and navigation scheme. Each of the following subsections details the purpose, workflow, features, database

interactions, validation, security, and machine learning integration of a specific system page, accompanied by a figure placeholder referencing the corresponding visual layout.

4.2.1 Landing Page

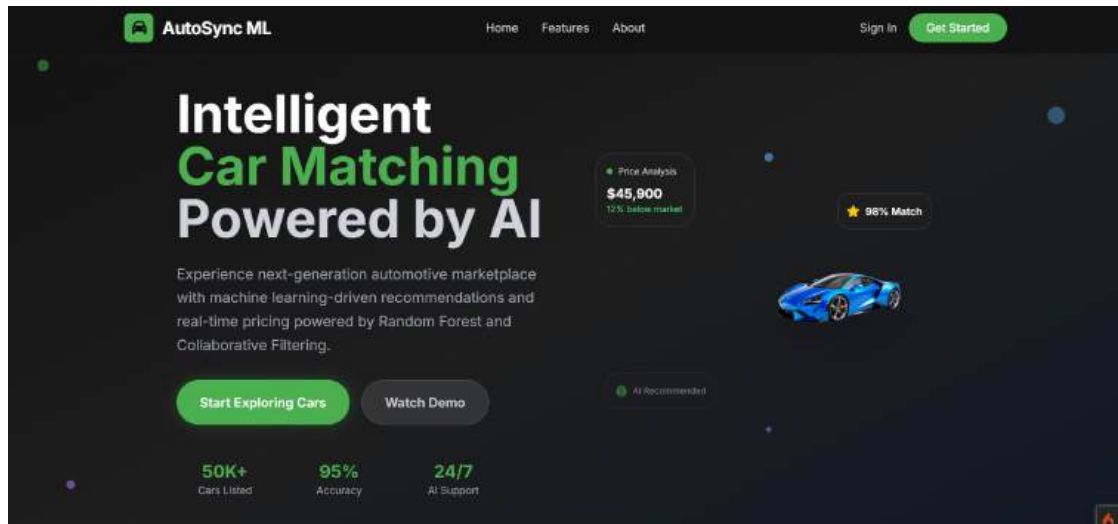


Figure 4.1: Landing Page

The landing page represents the primary interface encountered by unauthenticated visitors when accessing the application domain. The main purpose of this page is to introduce the platform's core functionalities, present a high-level overview of the marketplace inventory, and direct users toward onboarding or authentication pathways. The user workflow initiates with the loading of the main landing URL, which routes the request through Laravel's web routing engine to the landing page controller.

As illustrated in Figure 4.1, the landing page features a prominent navigation bar at the top, which contains links to browse listings, view the pricing tools, and access the authentication controls. Directly below the navigation header, a search section allows users to perform basic keyword queries against the active car database. The central portion of the page displays a grid of featured cars and an AI-recommended vehicles preview, showing a selection of high-demand vehicles. Call-to-action sections are positioned strategically to prompt unregistered visitors to sign up or log in.

Navigation from the landing page is designed to be frictionless, allowing users to jump directly to detailed search filters, registration forms, or the login portal. Input validation on this page is restricted to the basic search input, ensuring that query

strings are sanitized of potentially harmful characters before database execution. The database interaction involves executing a limited SELECT query via Laravel's Eloquent ORM to retrieve active listings with the highest view counts or those recently marked as featured. No direct machine learning model execution occurs on this page; instead, the system references a cached list of recommended vehicles computed daily to minimize runtime database load. By providing clear entry routes and a preview of the system's capabilities, the landing page establishes initial user engagement and guides visitors smoothly into the system's core workflows.

4.2.2 Login Page

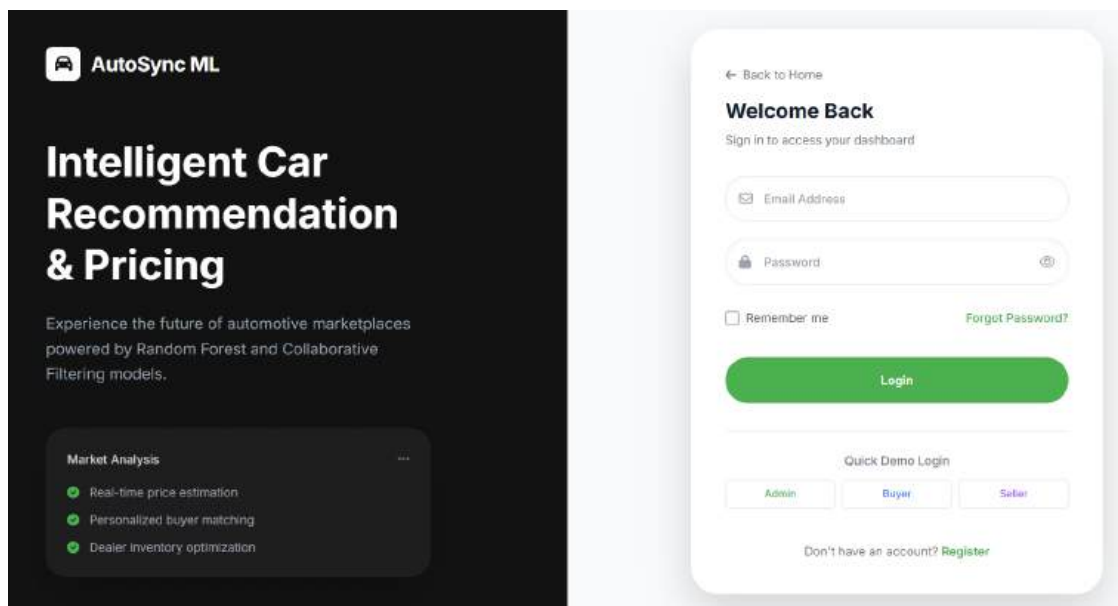


Figure 4.2: Login Page

The Login Page is the primary security gateway through which registered users authenticate their identity to access their personalized portals. The user workflow begins when a visitor clicks the login control on the navigation bar, redirecting them to the login form shown in Figure 4.2. The user enters their registered email address and password, optionally selecting the "Remember Me" checkbox to persist their session across browser restarts.

The main features of this page include the authentication credentials form, validation error message indicators, and links to the registration page and password recovery

utility. Upon clicking the submission button, client-side validation built into the Bootstrap form structure immediately checks that the email format is correct and that the password field is not empty. When submitted, Laravel's authentication controller intercepts the POST request and applies server-side validation to enforce request formatting constraints. The database interaction consists of querying the users table to match the email address. If a match is found, the system utilizes PHP's secure hashing functions to compare the input password against the bcrypt hash stored in the database.

Security features are heavily integrated into this page. Laravel's built-in Cross-Site Request Forgery (CSRF) protection is enforced using a hidden token field, preventing malicious external form submissions. If authentication succeeds, the controller checks the user's role flag (Buyer, Seller, or Admin) retrieved from the user record, writes the session variables, and issues a redirect response to the corresponding dashboard. If the credentials are invalid, the controller increments a throttling counter to prevent brute-force attacks and returns the user to the login page with an error alert. The implementation of this page ensures that user accounts are protected while directing authenticated users to their correct workspace.

4.2.3 Register Page

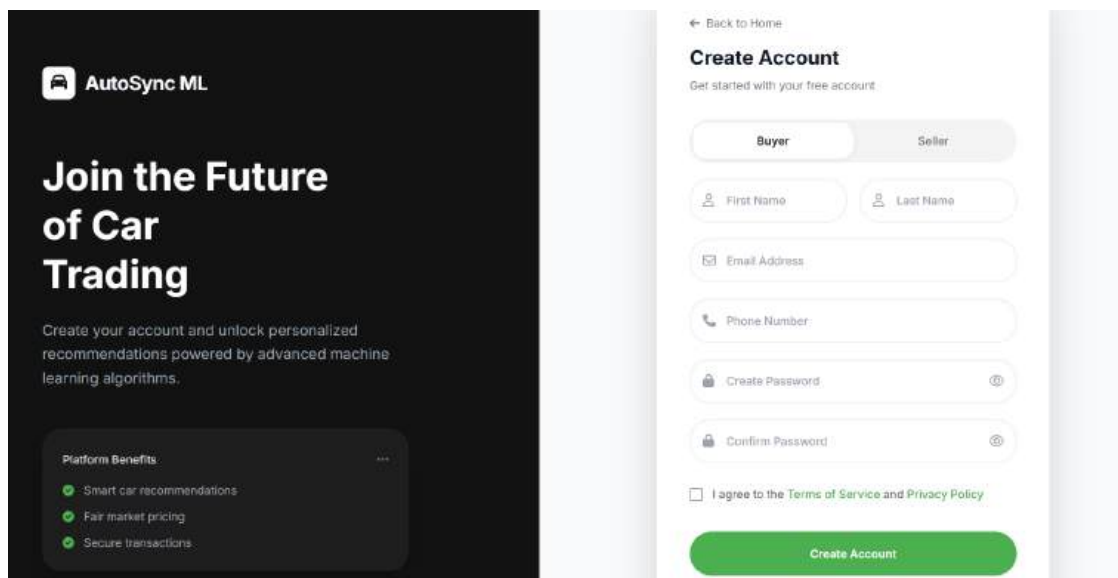


Figure 4.3: Register Page

The Register Page allows prospective buyers and sellers to create an account on the platform, establishing their profile and role within the database. The workflow begins

when a user navigates to the registration route, which displays the registration form shown in Figure 4.3. The user is required to select their account type—either Buyer or Seller—and fill in their full name, email address, phone number, password, and password confirmation.

The interface features custom dropdowns and text inputs styled with Bootstrap classes. The client-side validation logic verifies that all mandatory fields are populated, that the email address contains an "@" symbol, and that the password is at least eight characters long and matches the confirmation input. Upon clicking the register button, the form is posted to the registration controller. The controller executes server-side validation rules, checking the uniqueness of the email address against the database. If validation fails, the user is redirected back with input data preserved and validation errors highlighted.

The database interaction involves inserting a new record into the users table via Eloquent. The password is encrypted using the bcrypt hashing algorithm prior to insertion, ensuring that plaintext passwords are never written to the database. Upon successful insertion, the controller creates an active session for the new user, triggers an optional verification email, and redirects the user to their respective dashboard. The structured workflow of the registration page ensures that user data is validated and stored securely, providing a smooth onboarding process for new marketplace participants.

4.2.5 Buyer Dashboard

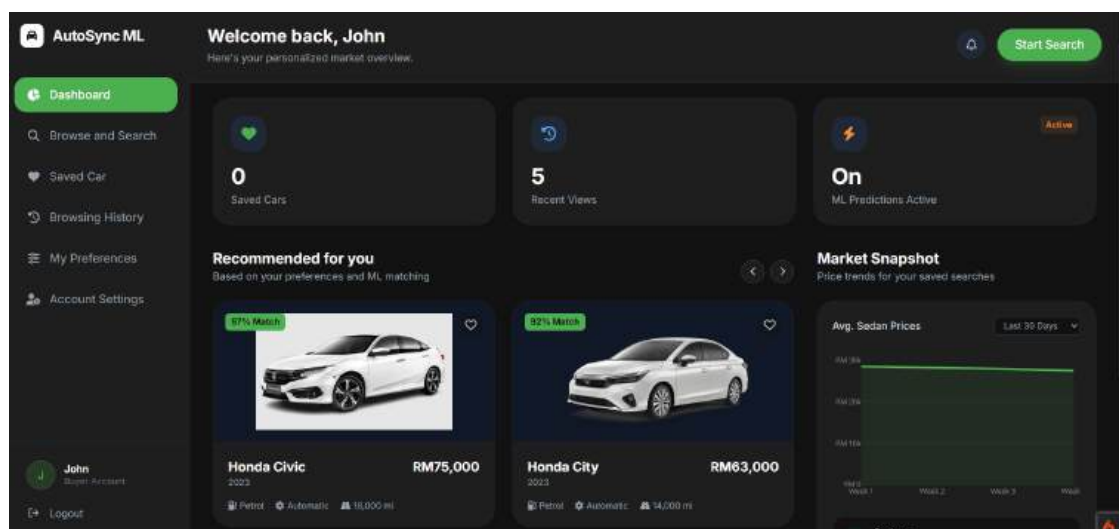


Figure 4.5: Buyer Dashboard

The Buyer Dashboard serves as the centralized dashboard for authenticated buyers, providing immediate access to personal marketplace activities and recommendation features. The workflow begins immediately post-login, routing the buyer to the dashboard interface depicted in Figure 4.5. The page layout consists of a persistent navigation header, a left-hand navigation menu, and a main content area.

The main features of this page are the activity summary cards, personalized recommendation panels, and a recent activity log. The summary cards display real-time counts of the buyer's saved cars, active search alerts, and recent views. Below these statistics, the personalized recommendations section displays a set of car listings selected by the system's Collaborative Filtering engine, tailored to the buyer's historical interactions. The recent activity section displays a list of recently viewed vehicles, enabling buyers to resume their browsing sessions easily.

The database interaction on this page is extensive. The controller executes queries using Eloquent ORM to retrieve the count of saved cars from the `saved_cars` table, recent views from the `browsing_history` table, and active listings from the `cars` table. The machine learning functionality is integrated via a database call that retrieves pre-calculated user recommendation vectors based on the buyer's user ID. No input validation is required on initial load, but the page incorporates security controls that verify the user session and block unauthorized access to the dashboard. The Buyer Dashboard improves platform usability by organizing key buyer metrics and personalized vehicle suggestions into a single, accessible workspace.

4.2.6 Browse Cars Page

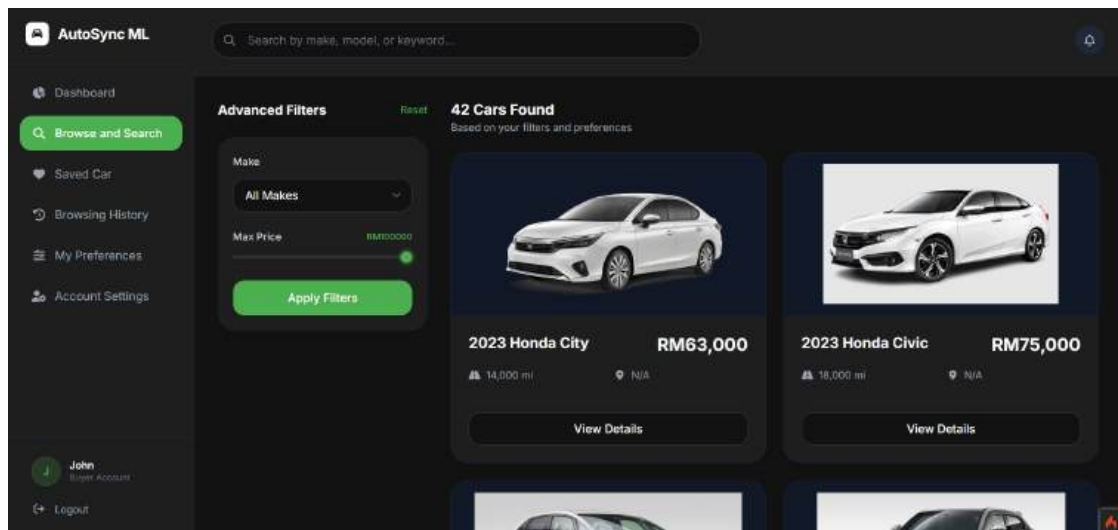


Figure 4.6: Browse Cars Page

The Browse Cars Page is the main search and exploration interface designed for buyers to discover listings within the marketplace. The workflow begins when the user clicks the "Browse" link in the navigation menu, landing on the interface shown in Figure 4.6. The page is structured into a sidebar containing filtering controls and a main grid displaying matching car listings.

The main features of this page include keyword search inputs, multidimensional filters (covering brand, model, price range, manufacturing year, fuel type, transmission, and body type), sorting dropdowns, and pagination controls. User interactions involve selecting filter parameters and clicking "Apply Filters," which triggers a GET request containing the query parameters. The browser page updates to display the filtered results. Input validation on the server side checks that minimum and maximum values are numeric and that the minimum price does not exceed the maximum price.

Database interaction is driven by a dynamic Eloquent query constructor. The controller evaluates the active query parameters and appends corresponding SQL WHERE clauses to the database query, restricting results to active listings matching the criteria. The sorting dropdown appends an orderBy clause, and Laravel's pagination method limits the results per page and renders the page links. Security is maintained by parameterizing all filter inputs, which prevents SQL injection attacks. This page enhances buyer usability by providing powerful filtering tools, allowing users to find specific vehicles quickly within a large database.

4.2.7 Car Details Page

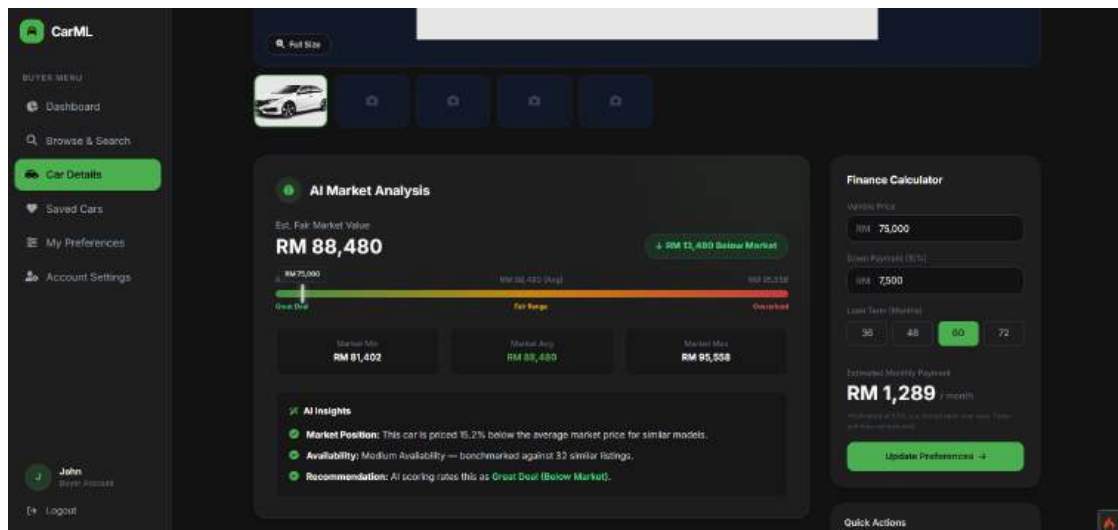


Figure 4.7: Car Details Page

The Car Details Page delivers a comprehensive breakdown of a single vehicle listing, displaying its complete technical specifications, image gallery, seller contact information, and similar recommended vehicles. The workflow begins when a buyer clicks on a car card from the landing page, browse page, or dashboard, loading the interface shown in Figure 4.7.

The primary features of this page include a tabbed information panel, an interactive photo viewer, a direct enquiry form, a "Save Car" button, and an AI-driven recommendations panel at the bottom of the page. The specifications section details the vehicle's engine capacity, mileage, fuel efficiency, transmission, and overall physical condition. The database interaction involves executing a SELECT query to retrieve the car record from the cars table, along with a JOIN query on the users table to fetch the listing seller's public contact details.

The machine learning functionality is integrated directly into this page. When the listing is retrieved, the system uses the car's attributes (such as brand, body type, and price) as input parameters for the Collaborative Filtering model. The system queries the database for other active listings that share high similarity scores with the current vehicle, rendering them in the "Similar Vehicles" panel. Security controls on this page check the active session to determine if the user has saved this vehicle, updating the "Save Car" button state accordingly. This interface improves buyer decision-making

by consolidating detailed specifications, seller contact details, and similar recommendations in one place.

4.2.8 Browsing History Page

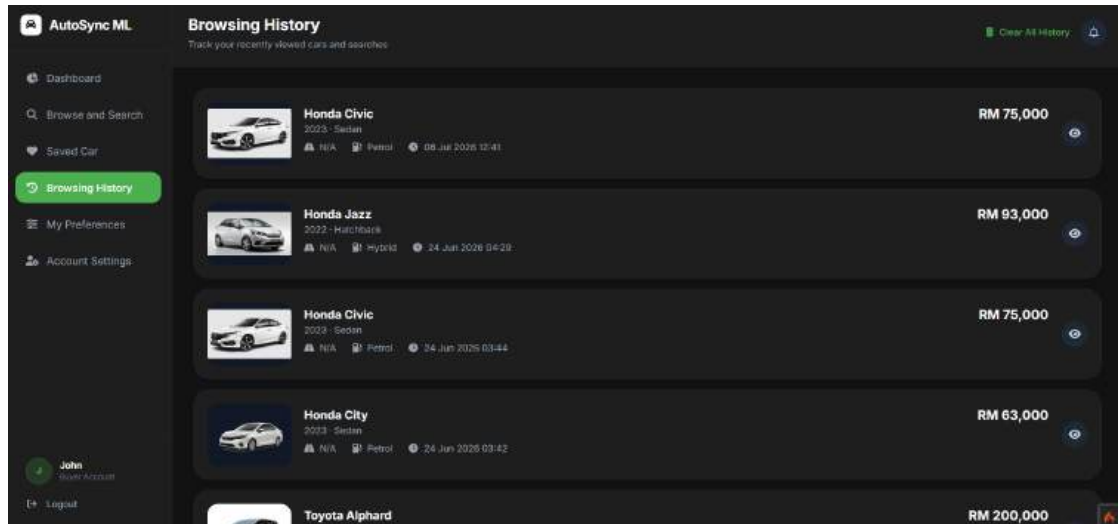


Figure 4.8: Browsing History Page

The Browsing History Page is a helper interface that allows buyers to track and revisit the car listings they have previously viewed. The user workflow initiates when a buyer accesses the history menu link, rendering the timeline view shown in Figure 4.8. The page displays a chronological list of recently viewed cars, with each entry showing the view timestamp and a link to the detail page.

The main features of this interface include the chronological history list, a search input to filter history entries, and a "Clear History" button. Each list entry displays a thumbnail image of the car, its brand and model name, the asking price, and the exact date and time the user viewed the page. Database interaction is handled via a middle-tier service that writes a log record to the `browsing_history` table whenever a user loads a car details page. When the history page is loaded, the controller executes a query to retrieve these records, joining them with the `cars` table to fetch vehicle details, sorted in descending order of the timestamp.

No complex input validation is required on this page, except when the user chooses to delete specific entries or clear their entire history, which triggers a delete query in the database. Security controls verify that the log entries displayed belong exclusively to the authenticated user's session ID. The browsing history page supports usability by

acting as a passive memory tool, allowing buyers to quickly locate and compare previously viewed vehicles without repeating search queries.

4.2.9 Car Preferences Page

The screenshot shows the 'My Preferences' page in the AutoSync ML application. The left sidebar contains navigation links: Dashboard, Browse and Search, Saved Car, Browsing History, My Preferences (highlighted), and Account Settings. The main content area is titled 'My Preferences' and includes a 'Buyer Profile' section with an 'Active Profile' button. The Buyer Profile section has fields for Budget Range (Rm 50,000.00 to Rm 500,000.00), Search Radius (Miles), Preferred Brand (Any Brand), Preferred Fuel Type (Petrol), and Preferred Body Type (Sedan). Below this is a 'Recommendation Weighting' section with sliders for Price & Value Score (High Priority) and Condition & Mileage (Medium Priority). To the right is an 'Alert Rules' section with toggles for Price Drops, New Matches, and Market Moving, and a 'Delivery Method' section (Email, Push, SMS). At the bottom right is a 'Privacy & Data' section with checkboxes for Share anonymized search data and Allow dealers to contact me, and a link to Clear Search History.

Figure 4.9: Car Preferences Page

The Car Preferences Page allows buyers to explicitly define their vehicle preferences to guide the system's recommendation engine. The workflow begins when the buyer navigates to the preference settings menu, loading the form shown in Figure 4.9. The page features selection controls for preferred brands, price range constraints, preferred body styles, transmission type, and fuel type.

The features of this interface include multi-select checkboxes for brands, numeric input fields for price ranges, and radio buttons for transmission types. Input validation rules require that the price inputs contain positive numeric values and that the maximum price is higher than the minimum price. Upon form submission, the controller updates the buyer's preference record in the `user_preferences` table. If no preference record exists, a new entry is created.

The machine learning functionality uses these stored preferences as a user preference vector. The Collaborative Filtering algorithm matches this vector against the item feature matrix to compute similarity scores for all active car listings. Listings that match the user's explicit preferences are given priority weightings in recommendation calculations. The resulting personalized recommendations are subsequently displayed on the buyer's dashboard. This page improves user satisfaction by giving buyers direct

control over the AI-driven recommendation logic, ensuring the suggested vehicles align with their specific preferences.

4.2.10 Saved Cars Page

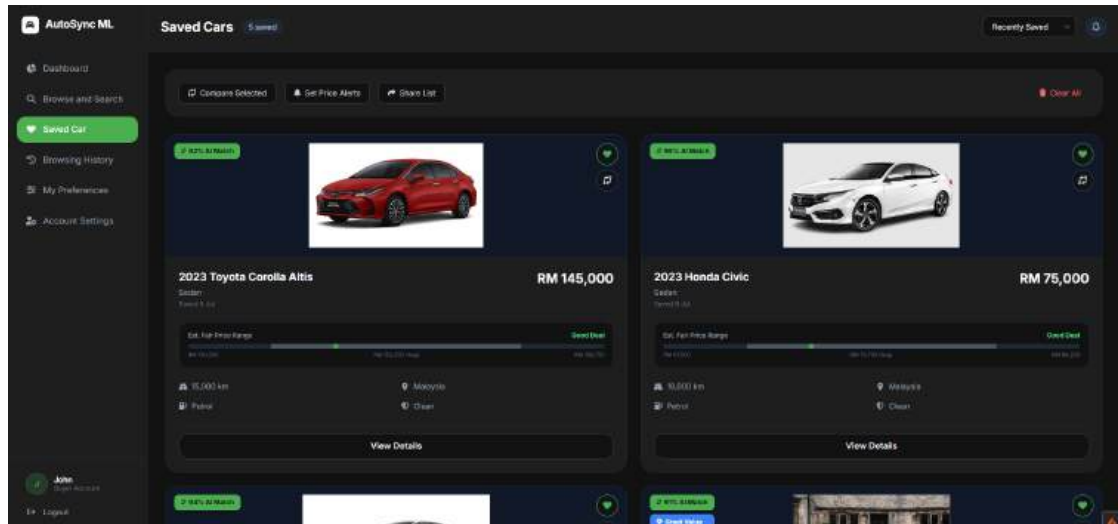


Figure 4.10: Saved Cars Page

The Saved Cars Page, commonly referred to as the wishlist, is a dedicated dashboard area where buyers can monitor and compare their bookmarked listings. The workflow begins when the user selects the "Saved Cars" menu option, loading the grid view shown in Figure 4.10. The page presents a clean layout containing cards of all currently saved vehicles.

The primary features of this page include the saved listings grid, a vehicle comparison tool launcher, and "Remove" controls on each card. The user workflow involves browsing these saved listings and optionally selecting up to three vehicles to compare side-by-side. Database interactions occur when the page loads; the controller queries the `saved_cars` table using the user's ID, performs a JOIN query on the `cars` table to retrieve active vehicle data, and checks for updates (such as price reductions or status changes).

Input validation is minimal and limited to verifying the listing ID when a user removes a car from their wishlist. The delete action triggers an AJAX request that deletes the matching record from the `saved_cars` table, updating the user interface dynamically. Security checks confirm that the user owns the wishlist record before executing the deletion. The Saved Cars Page improves usability by allowing buyers to

shortlist candidates and track price changes, helping them make more informed purchasing decisions.

4.2.11 Buyer Settings Page

Figure 4.11: Buyer Settings Page

The Buyer Settings Page provides buyers with the interface required to manage their account profile and security credentials. The user workflow begins when the buyer navigates to the account settings section, loading the interface shown in Figure 4.11. The page is split into two primary forms: Profile Information and Change Password.

The main features of this page include text input fields for the user's name, email address, phone number, and password updates. Input validation on this page is strict. The profile update form verifies that the name field is not empty, that the email conforms to standard formatting, and that the email is not already in use by another account. The password update form requires the user to enter their current password, their new password, and a password confirmation, validating that the new password is at least eight characters long.

Database interaction involves executing an SQL UPDATE query on the users table to save the updated profile fields. If a password change is requested, the controller verifies the current password using `password_verify()`. Once verified, the new password is encrypted using the `bcrypt` algorithm and updated in the database. Security is maintained by requiring session re-authentication for sensitive modifications. This page is essential for account security, giving buyers control over their personal data and authentication credentials.

4.2.12 Seller Dashboard

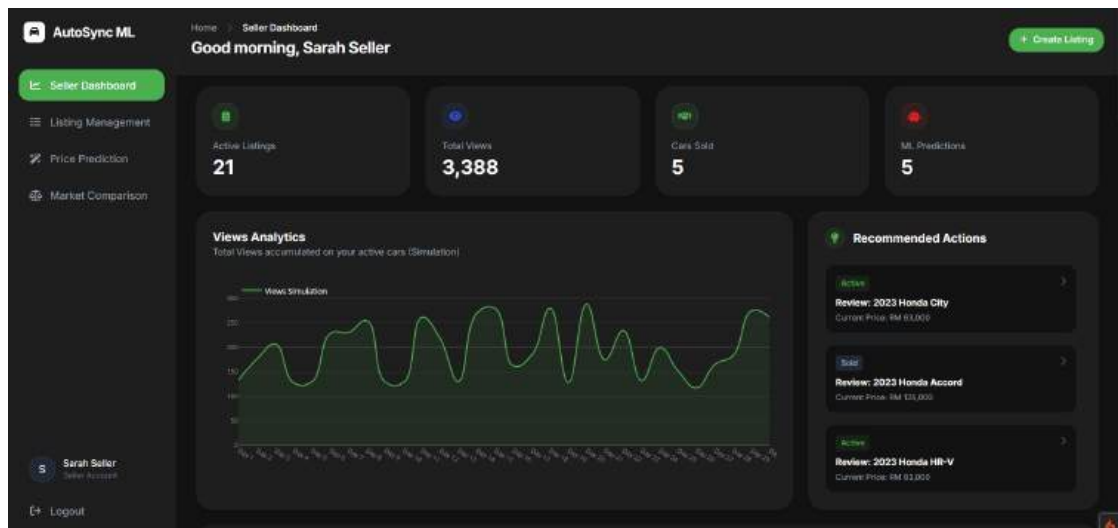


Figure 4.12: Seller Dashboard

The Seller Dashboard is the main administration panel for registered sellers, designed to summarize listing performance and provide quick access to inventory management. The workflow begins post-login when a seller is redirected to the interface shown in Figure 4.12. The dashboard uses Bootstrap containers to organize statistical panels and lists.

The key features of this dashboard are listing performance metrics, a pricing insight summary, and recent notifications. The metrics panel displays the seller's total active listings, overall listing views, and customer inquiries. The pricing insight card provides a summary of the seller's listings that are priced above or below the market average. Database interactions involve executing query counts on the cars table filtered by the seller's user ID, alongside aggregate queries on a listing_views table to sum views across all listings owned by the seller.

No input validation occurs on the initial dashboard load, but access control is managed via route middleware that checks for active seller sessions. The dashboard integrates machine learning outputs by displaying a summary of the Random Forest model's valuations for the seller's active listings, highlighting listings that may benefit from price adjustments. This dashboard improves seller usability by consolidating listing views, active inventory status, and pricing insights into a single view.

4.2.13 Add Listing Page

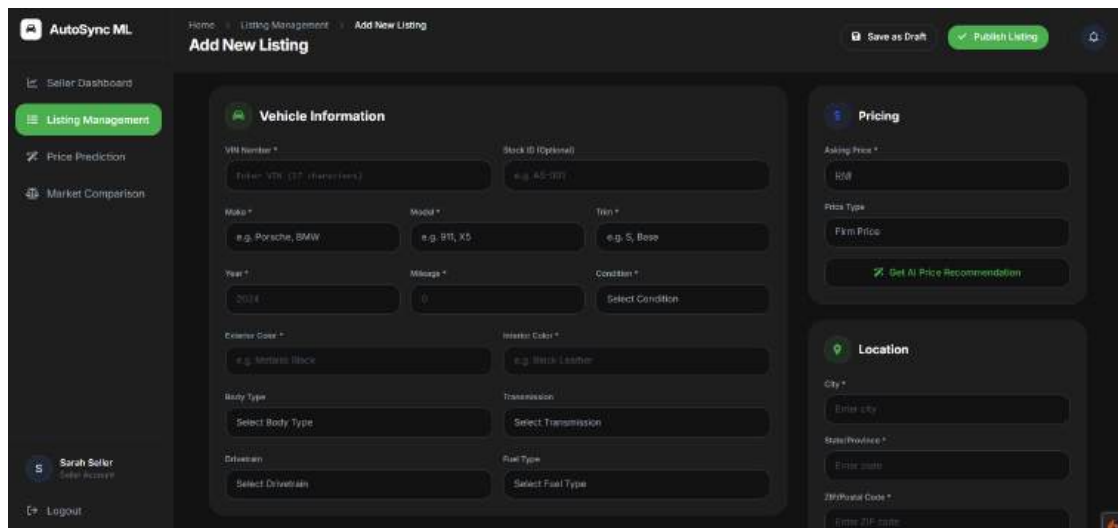


Figure 4.13: Add Listing Page

The Add Listing Page is the input interface through which sellers publish new vehicle listings to the marketplace. The workflow begins when the seller clicks the "Add New Listing" button on their dashboard, loading the form shown in Figure 4.13. The form is divided into sections: Basic Information, Technical Specifications, and Media Upload.

The key features of this page are the input inputs (including dropdowns for brand and model, text fields for mileage, price, engine size, fuel type, and transmission), a text area for descriptions, and a file drag-and-drop zone for uploading vehicle images. Input validation is applied on this page to ensure listing quality. Client-side validation checks that fields like price and mileage contain positive numeric values, that the model year is a valid four-digit integer, and that at least one primary image is selected.

Upon submission, the data is processed by the listing controller. The controller validates that the image files do not exceed size limits and conform to allowed formats (JPEG, PNG, WebP). The database interaction involves running an INSERT query on the cars table to create the listing record, referencing the seller's user ID as the foreign key. Image files are saved to the server's public storage directory, and their file paths are saved to the database. This page is critical to the marketplace workflow, providing a structured form to ensure all listings are populated with complete and accurate information.

4.2.14 Edit Listing Page

The screenshot shows the 'Edit Listing' page for a 2023 Honda City. The interface is dark-themed. On the left is a sidebar with a user profile for 'Sarah Seller' and navigation links. The main content area has a top navigation bar with 'Home', 'Listing Management', and 'Edit Listing'. Below this, the form is pre-filled with the following details:

- Make:** Honda
- Model:** City
- Year:** 2023
- Mileage:** 14000
- Condition:** Excellent
- Exterior Color:** Dolomite Silver Metallic
- Interior Color:** Black Leather

The 'Pricing & Market Analysis' section displays:

- Your Asking Price:** RM 63,000.00
- Price Type:** Firm Price
- AI Market Analysis:** Market Average RM 83,952 (1.0% below), Your Position RM 63,000 (Competitive), Est. Days to Sell 15-25 days.

The 'Listing Performance' section shows:

- Total Views (24h):** 1,245
- Leads Generated:** 42

At the top right, there are buttons for 'Preview Listing' and 'Save Changes'.

Figure 4.14: Edit Listing Page

The Edit Listing Page allows sellers to modify the details of their active car listings. The workflow begins when a seller clicks the "Edit" button on a listing from their dashboard or management list, loading the interface shown in Figure 4.14. The page displays the listing creation form pre-populated with the vehicle's current database values.

The main features of this page are the pre-filled inputs, the current image preview panel with delete controls, and the form submission buttons. The database interaction begins with a SELECT query on the cars table using the listing ID to load the current details into the form inputs. Security verification is performed in the controller to confirm that the seller_id stored on the listing matches the authenticated user's ID, preventing unauthorized modification of other users' listings.

The seller can modify specifications, update the text description, delete existing photos, or upload new images. Input validation on submission is identical to the listing creation validation, checking numeric values, required fields, and image formats. Upon validation, the controller executes an SQL UPDATE query on the cars table. Modified image paths are updated, and deleted images are removed from the server storage. The Edit Listing Page provides sellers with an efficient tool for maintaining listing accuracy, ensuring changes are saved securely back to the database.

4.2.15 Manage Listings Page

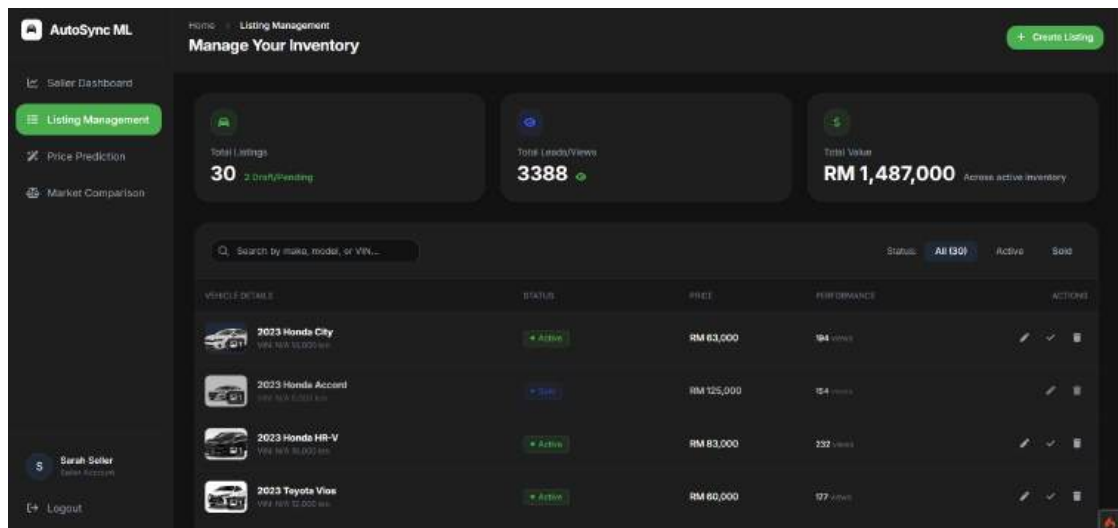


Figure 4.15: Manage Listings Page

The Manage Listings Page is the primary inventory management panel for sellers, providing an overview of all their listings. The user workflow begins when the seller selects the "Manage Listings" menu item, loading the interface shown in Figure 4.15. The page displays a table containing all active and inactive listings created by the seller.

The main features of this interface include a tabular inventory list, status filters (All, Active, Pending, Sold), a keyword search bar, and action buttons (Edit, Delete, Mark as Sold) for each listing. The database interaction involves querying the cars table, filtering the records by the seller's user ID, and displaying them with pagination. The list displays the primary image thumbnail, vehicle title, asking price, current status, and accumulated page views.

When a seller initiates a deletion, client-side validation triggers a confirmation modal. If confirmed, a POST request is sent to the deletion controller. The controller verifies the user's ownership of the listing, deletes the record from the cars table, and removes the associated images from the server storage. This page improves seller workflow usability by providing a structured view to monitor, edit, and delete listings from a single page.

4.2.16 Market Comparison Page

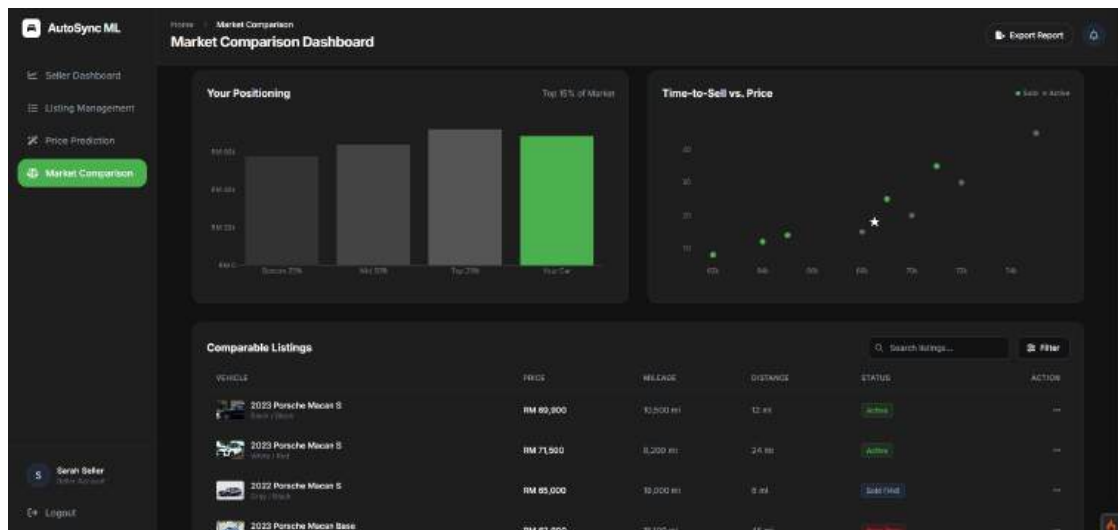


Figure 4.16: Market Comparison Page

The Market Comparison Page is an analytical interface that helps sellers evaluate their listings against similar vehicles in the marketplace. The workflow begins when a seller selects a listing to analyze, loading the comparative interface shown in Figure 4.16. The page is designed with progress bars, statistics cards, and price range charts.

The main features of this page are the listing comparison widget, price summary cards (displaying the highest, lowest, and average prices of similar models), and a market positioning indicator. Database interaction involves executing SELECT queries with aggregation functions (MAX, MIN, AVG) on the cars table, filtering by matching criteria like brand, model, manufacturing year, and transmission type.

This page integrates machine learning by calculating the difference between the seller's listing price and the model's predicted valuation. The interface renders a visual indicator showing where the listing sits relative to the market distribution (e.g., competitive, average, or overpriced). This tool benefits sellers by providing a clear view of their pricing relative to the current market, helping them set competitive prices that attract buyers while maximizing returns.

4.2.17 Price Prediction Page

Figure 4.17: Price Prediction Page

The Price Prediction Page is the primary interface for the system's machine learning price prediction subsystem, allowing sellers to estimate their vehicle's value. The workflow starts when a seller loads the page, inputting their car's specifications into the prediction form shown in Figure 4.17.

The key features of this page are the vehicle parameters form (brand, model, manufacturing year, mileage, engine size, fuel type, transmission, and overall condition) and the prediction results panel. Input validation checks that all required fields are filled, that mileage and engine size are positive numbers, and that the year is a valid four-digit integer. Upon submission, the form is processed by the pricing controller, which passes the inputs to the Random Forest model.

The prediction workflow executes the serialized Random Forest model using the input features. The script generates an estimated market price, returning it to the controller. The database is queried to find similar listings to display alongside the estimate. The results panel renders the predicted price in Malaysian Ringgit (MYR), along with a price range indicating typical market variance. This tool assists sellers by providing data-driven price estimates, removing guesswork from listing pricing, and improving listing competitiveness.

4.2.18 Admin Dashboard

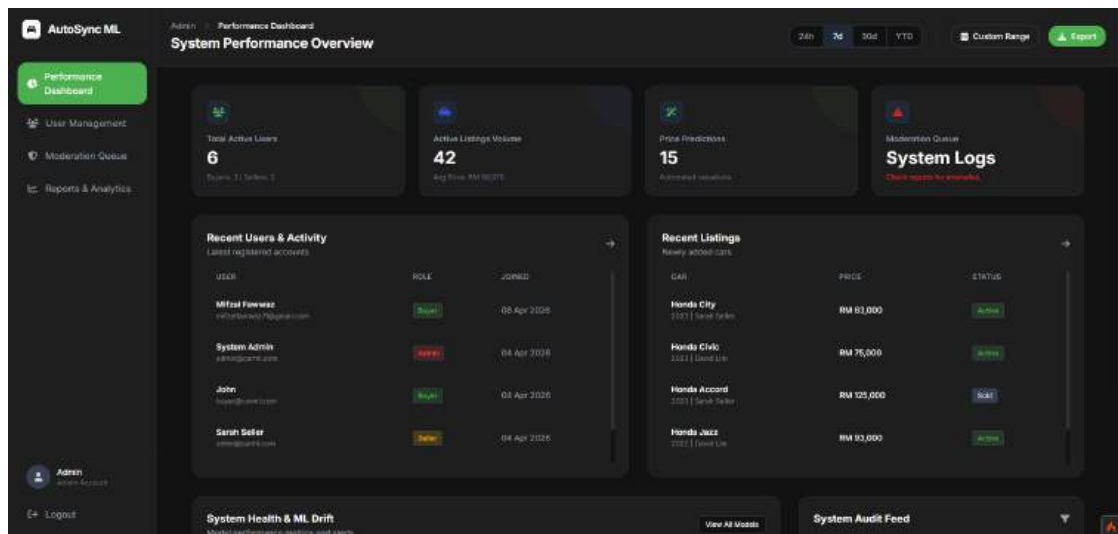


Figure 4.18: Admin Dashboard

The Admin Dashboard is the central command console for system administrators, providing an overview of platform activity and system health. The workflow begins when an administrator authenticates and routes to `/admin/dashboard`, loading the interface shown in Figure 4.18.

The features of this dashboard are system status metrics cards, listing moderation tables, user registration charts, and system logging reports. The metrics cards display total users, active listings, and machine learning statistics (prediction counts and recommendation system activity). Database interactions involve executing count queries across the `users`, `cars`, `saved_cars`, and `search_queries` tables.

Security is managed via strict role-based access control (RBAC) middleware, rejecting non-admin access attempts. The page displays the latest pending listings, allowing the admin to approve or reject them directly. The Admin Dashboard improves usability by consolidating system-wide metrics, active listing alerts, and user registrations into a single administrative view.

4.2.19 Manage Listings Page (Admin)

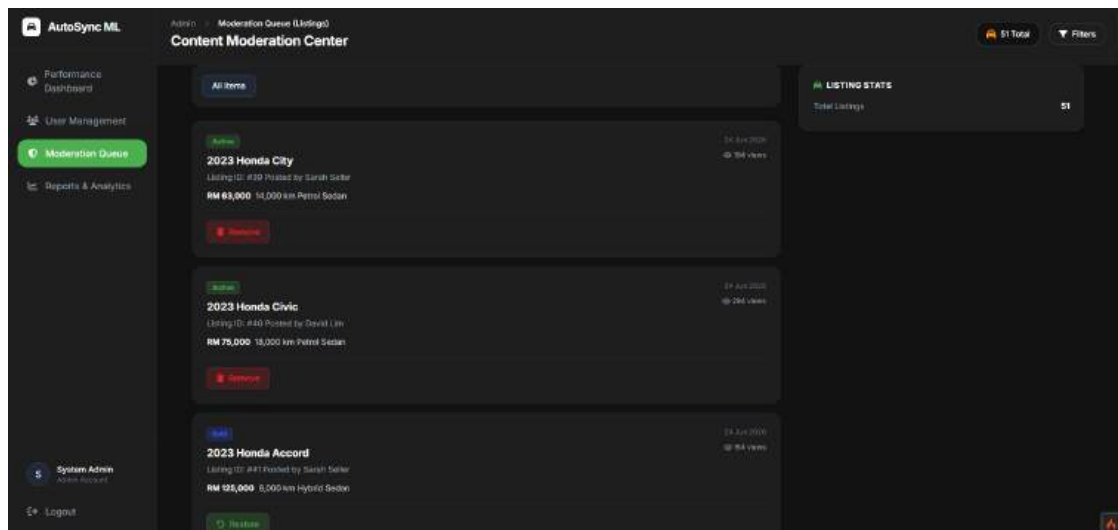


Figure 4.19: Manage Listings Page (Admin)

The Admin Manage Listings Page gives administrators oversight of all listings on the platform. The workflow begins when the admin navigates to the listing management portal, loading the interface shown in Figure 4.19.

The main features of this page are the platform-wide listings table, status filters (Active, Pending, Rejected, Flagged), search controls, and moderation actions (Approve, Reject, Delete). Database interaction involves querying the cars table to retrieve all listings, joining with the users table to fetch owner information.

When an admin clicks "Reject," a modal prompts them to input a reason. This reason is saved, the listing status is updated to "Rejected," and the seller is notified. If a listing is flagged for policy violations, the admin can delete it, which triggers a delete query on the cars table and removes the listing from search results. This page is essential for content moderation, ensuring listings comply with platform standards and resolving reports of fraudulent listings.

4.2.20 Manage Users Page (Admin)

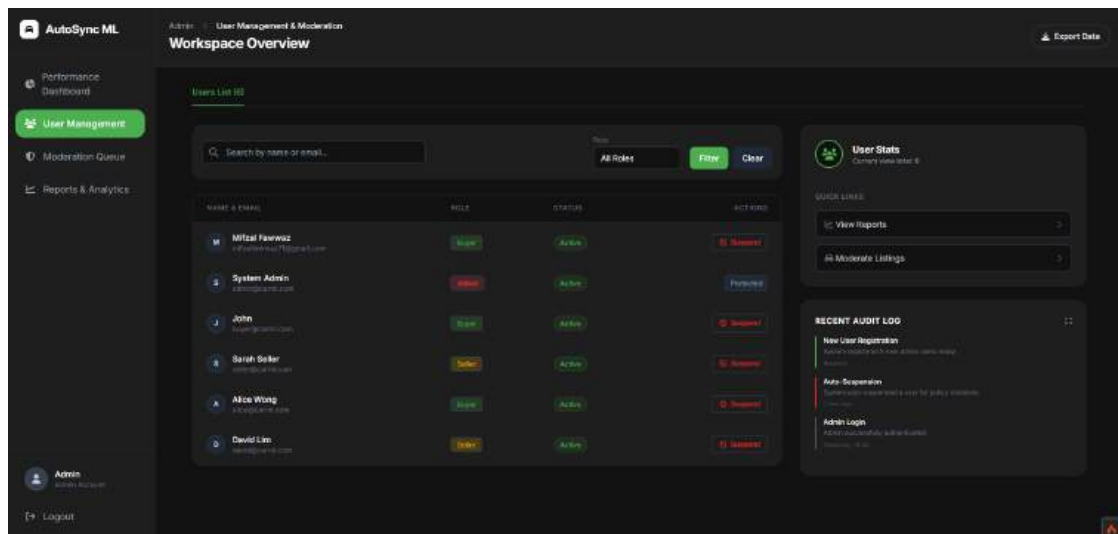


Figure 4.20: Manage Users Page (Admin)

The Manage Users Page provides administrators with the controls required to manage user accounts on the platform. The workflow begins when the admin navigates to the user management portal, loading the interface shown in Figure 4.20.

The primary features of this page are the registered users table, role filters, user status controls (Active, Suspended), a user search input, and profile details panels. Database interactions involve executing SELECT queries on the users table, filtering by search keywords, and displaying the results in a paginated list.

Administrators can toggle a user's status between "Active" and "Suspended." The suspend action updates the user's status in the database, blocking their session and login access. The admin can also delete accounts, which removes the user's record from the users table. The Manage Users Page is critical for platform governance, allowing the administrator to manage user accounts and take action against bad actors.

4.2.21 Reports Page



Figure 4.21: Reports Page

The Reports Page is the analytical dashboard for administrators to evaluate platform trends and user engagement. The workflow begins when the admin accesses the reports section, loading the interface shown in Figure 4.21.

The primary features of this page are reports on sales volume, listing growth, active users, recommendation performance, and price prediction usage. Database interactions involve running aggregation and grouping queries on tables like cars, users, saved_cars, and prediction_logs over weekly, monthly, or yearly periods.

The controller processes these query results into structured arrays, rendering them as charts and exportable tables (supporting CSV and PDF formats). These analytics help the administrator monitor system metrics, identify growth trends, and evaluate the performance of the machine learning modules, assisting in platform planning.

4.4 Testing Analysis

Testing is a critical phase of system development that validates the implementation against the project requirements. The system underwent testing using a combination of black-box and white-box methodologies to evaluate functional completeness, security, and performance.

4.4.1 Black-Box Testing

Black-box testing, also known as functional testing, evaluates the system's external behavior without reviewing the underlying source code or database queries. The tester

interacts with the user interface as an end user, verifying that inputs generate the expected outputs and that navigation flows correctly.

For this project, black-box testing was conducted on the following key modules:

Authentication (Login, Registration, Forgot Password): Verified that user accounts could be created, credentials validated, and sessions terminated correctly.

Buyer Navigation (Dashboard, Browse Cars, Car Details, Browsing History, Saved Cars, Preferences): Confirmed that buyers could browse listings, save vehicles to wishlists, view similar recommendations, and set search preferences.

Seller Inventory (Add Listing, Edit Listing, Manage Listings, Market Comparison, Price Prediction): Evaluated listing creation and editing, image upload handling, market analysis metrics, and valuation calculations.

Administrative Portal (Dashboard, User Management, Listing Moderation, System Reports): Tested the administrative interfaces, confirming that the admin could moderate listings, manage user accounts, and view platform reports.

Each black-box test was documented in structural tables, detailing the steps, expected results, and actual outcomes to confirm functional compliance.

4.4.2 White-Box Testing

White-box testing examines the internal structure and logic of the source code. It requires access to the codebase to verify components like execution paths, database queries, security controls, and exception handling.

The white-box testing process for this system focused on:

- **Laravel Controllers & Models:** Reviewed controller code to ensure validation rules were applied, input parameters were handled correctly, and model methods returned the expected data objects.
- **Routing & Middleware:** Inspected route definitions and middleware filters (such as authentication and role authorization checks) to confirm that unauthorized users were blocked from protected areas.

- **Database Queries:** Audited Eloquent queries and raw SQL statements to ensure they used parameterization, protecting the database against SQL injection.
- **Machine Learning Integration Layer:** Tested the system process calls to the Python scripts to verify that execution parameters were passed correctly, script errors were handled, and the JSON outputs were parsed without throwing exceptions.
- **Security & Validation Controls:** Inspected security implementations, including CSRF token verification on forms and input validation parameters on user forms.

White-box testing helped identify logic errors, verify access control boundaries, and ensure the backend components integrated correctly with the front-end user interface.

4.5 Test Cases

This section contains the software testing tables detailing the test cases executed to validate the platform. The tables are grouped by system module: General (Authentication & Landing Page), Buyer, Seller, and Administrator.

4.5.1 General Module Test Cases

The table below outlines the functional test cases executed against the general and authentication interfaces of the platform.

Table 4.5.1 General Module Test Case

Step	Procedure	Expected Result	Pass/Fail
1	Access landing page URL. Observe search, featured grid, navigation, and register links.	Page renders, showing the landing page components, navigation headers, and call-to-actions.	Pass

2	On landing page, input search query "Toyota" and submit form.	System routes search request to Browse page, pre-filtering results to display matching listings.	Pass
3	Navigate to login page, enter registered email and valid password. Click "Sign In".	User session is established; user is redirected to their role's dashboard.	Pass
4	Navigate to login page, enter registered email and invalid password. Click "Sign In".	Login fails, an error message is displayed, and the session is not created.	Pass
5	Navigate to login page, enter unregistered email and click "Sign In".	Login fails, and a generic credential error message is displayed to prevent user enumeration.	Pass
6	Navigate to registration page, fill in valid user details, select account type, and click "Submit".	A new record is added to the users table with a hashed password, and the user is redirected.	Pass
7	Navigate to registration page, enter an email already in use. Click "Submit".	Registration fails, and a validation message is displayed indicating the email is taken.	Pass

8	Navigate to forgot password page, enter a registered email, and click "Reset".	Reset token is stored in the database, and a password reset link is sent to the email.	Pass
9	Click an expired reset link from email.	Reset attempt is rejected, and the system redirects to the forgot password page with an error.	Pass

4.5.2 Buyer Module Test Cases

The table below presents the test cases for the buyer interfaces, including navigation, browse queries, wishlist management, and recommendations.

Table 4.5.2 Buyer Module Test Cases

Step	Procedure	Expected Result	Pass/Fail
1	Log in as buyer, access dashboard. Review statistics cards and recommendation panels.	Dashboard loads, displaying the user's statistics and personalized recommendations.	Pass
2	Navigate to browse page. Set price range filter to 50,000-80,000 and click "Apply".	Listings grid updates, showing only vehicles with prices within the selected range.	Pass
3	Navigate to browse page. Apply brand filter "Honda" and body style "Sedan".	Listings grid updates to show only Honda sedans.	Pass

4	Click on a listing card. Review details, specifications, seller details, and recommendations.	Details page loads, showing the technical specifications, seller info, and similar recommendations.	Pass
5	On details page, click "Save Car" button.	Listing is saved to the wishlist. The button updates to reflect the saved state.	Pass
6	Access Saved Cars page. Review wishlisted listings.	Wishlist page loads, displaying the previously saved car listings.	Pass
7	Click "Remove" button on a saved car card.	Car is removed from the wishlist, and the listing is removed from the page.	Pass
8	Access Preferences page. Select preferred brands and price range. Click "Save".	Preferences are saved to the database. The dashboard updates to display matching recommendations.	Pass
9	Access Browsing History page. Verify recently viewed cars list.	Timeline list loads, showing the previously viewed cars in chronological order.	Pass
10	Access Profile settings page. Update name, phone, and password details. Click "Update".	Profile values are updated in the database, and a success message is displayed.	Pass

4.5.3 Seller Module Test Cases

The table below details the functional test cases executed to validate the seller-specific features.

Table 4.5.3 Seller Module Test Cases

Step	Procedure	Expected Result	Pass/Fail
1	Log in as seller. Access dashboard to review listing stats and valuation summaries.	Dashboard loads, displaying the listing metrics and valuation summaries.	Pass
2	Access Add Listing page. Fill in details, select a image, and click "Submit".	Listing is created in the database, image is saved, and the listing is displayed in the marketplace.	Pass
3	Access Add Listing page. Submit form with empty fields.	Form submission is blocked, and validation messages are displayed for the missing inputs.	Pass
4	Access Edit Listing page for an owned car. Update the price and click "Save Changes".	Listing record is updated in the database, and a success message is displayed.	Pass
5	Attempt to access the edit listing URL of a car owned by a different seller.	Request is blocked by authorization middleware, returning a 403 error page.	Pass

6	Access Manage Listings page. Click "Delete" on a listing and confirm the prompt.	Listing is deleted from the database, images are deleted, and the list updates.	Pass
7	Access Price Prediction page. Input vehicle specifications and click "Predict".	Random Forest model processes the input variables, and the estimated price is displayed.	Pass
8	Access Market Comparison page for a listing. Review pricing metrics.	Comparison page loads, showing similar listings, price averages, and market comparison metrics.	Pass

4.5.4 Administrator Module Test Cases

The table below details the administrative test cases for platform governance, moderation, and reporting.

Table 4.5.4 Administrator Module Test Cases

Step	Procedure	Expected Result	Pass/Fail
1	Log in as administrator. Access admin dashboard to check system statistics.	Dashboard loads, displaying total users, active listings, and machine learning logs.	Pass
2	Access Manage Listings page. Search for listing "Toyota" using query input.	Platform-wide listings table filters, showing only listings containing "Toyota".	Pass

3	On Manage Listings page, select a pending listing and click "Approve".	Listing status is updated to "Active" in the database, and it is displayed in search results.	Pass
4	On Manage Listings page, select a listing and click "Delete". Confirm deletion.	Listing is deleted from the database, images are deleted, and listing is removed from search.	Pass
5	Access Manage Users page. Search for user by email address.	Users table filters, displaying the user matching the email address.	Pass
6	On Manage Users page, click "Suspend" on a seller's user row.	User's status is updated to "Suspended" in the database. The user is blocked from logging in.	Pass
7	Access Reports page. Select date range and click "Generate".	Report is generated, displaying sales metrics, registration growth, and system usage.	Pass
8	Click "Export to CSV" on a generated report page.	Report data is compiled and downloaded as a formatted CSV file.	Pass

CHAPTER 5

CONCLUSION

5.1 Introduction

This chapter discusses how the developed **Intelligent Car Recommendation and Pricing System Using Machine Learning** contributes to improving decision-making in online car marketplaces. The system assists buyers by providing personalised vehicle recommendations while supporting sellers with data-driven car price prediction. This chapter also discusses the limitations of the developed system and proposes several recommendations for future improvements to enhance its functionality, accuracy, and usability.

5.2 Contribution

The **Intelligent Car Recommendation and Pricing System Using Machine Learning** was developed to improve the efficiency of online car buying and selling by integrating machine learning techniques into a web-based platform. The system combines collaborative filtering for personalised car recommendations and regression-based price prediction to support better decision-making for both buyers and sellers.

For buyers, the recommendation module reduces the time required to search for suitable vehicles by suggesting cars based on browsing history, preferences, and previous interactions. The predicted market price also allows buyers to evaluate whether a vehicle is reasonably priced before making a purchasing decision.

For sellers, the system provides an intelligent price estimation feature that generates suggested selling prices based on important vehicle attributes such as brand, model, manufacturing year, mileage, and engine specifications. This enables sellers to determine competitive and realistic prices using data-driven analysis rather than relying solely on personal judgement.

From the administrator perspective, the system provides facilities to manage user accounts, monitor car listings, validate system data, and oversee overall platform

activities. The integration of machine learning into a single online car marketplace demonstrates the practical application of artificial intelligence in automotive e-commerce while improving pricing transparency, recommendation accuracy, and user experience.

5.3 Limitation of Work

Although the developed system provides intelligent recommendation and price prediction capabilities, several limitations remain.

Firstly, the recommendation quality depends heavily on the availability of user interaction data. Newly registered users who have little or no browsing history may receive less accurate recommendations due to the cold start problem commonly found in collaborative filtering algorithms.

Secondly, the price prediction model estimates vehicle prices using historical vehicle attributes and available datasets. External factors such as sudden market fluctuations, economic conditions, vehicle demand, seasonal trends, negotiation strategies, and government policies are not fully considered. Therefore, the predicted prices may differ from actual market values under changing market conditions.

Lastly, the system is implemented as a web-based application and has been evaluated within a controlled development environment. Performance under large-scale deployment involving thousands of concurrent users has not been extensively tested. Therefore, scalability, response time, and overall system performance may vary in real-world environments.

5.4 Future Works

Several enhancements can be implemented in future studies to further improve the performance and functionality of the proposed system.

Firstly, a hybrid recommendation approach that combines collaborative filtering with content-based filtering can be implemented to reduce the cold start problem and improve recommendation accuracy for new users who have limited interaction history.

Secondly, the price prediction model can be enhanced by incorporating real-time market information collected from external automotive marketplaces and APIs. Additional variables such as regional demand, fuel prices, economic indicators, vehicle accident history, and maintenance records can also be included to improve prediction accuracy.

Furthermore, future versions of the system may support mobile application development to provide users with easier access through smartphones and tablets. Additional features such as online vehicle booking, secure payment integration, vehicle inspection reports, seller verification, and live communication between buyers and sellers can further improve the overall user experience.

Finally, more advanced machine learning and deep learning algorithms can be evaluated and compared with the existing models to improve recommendation relevance and pricing accuracy. Continuous model retraining using larger and more diverse datasets would allow the system to adapt to changing market trends and provide more reliable recommendations and price predictions over time.

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